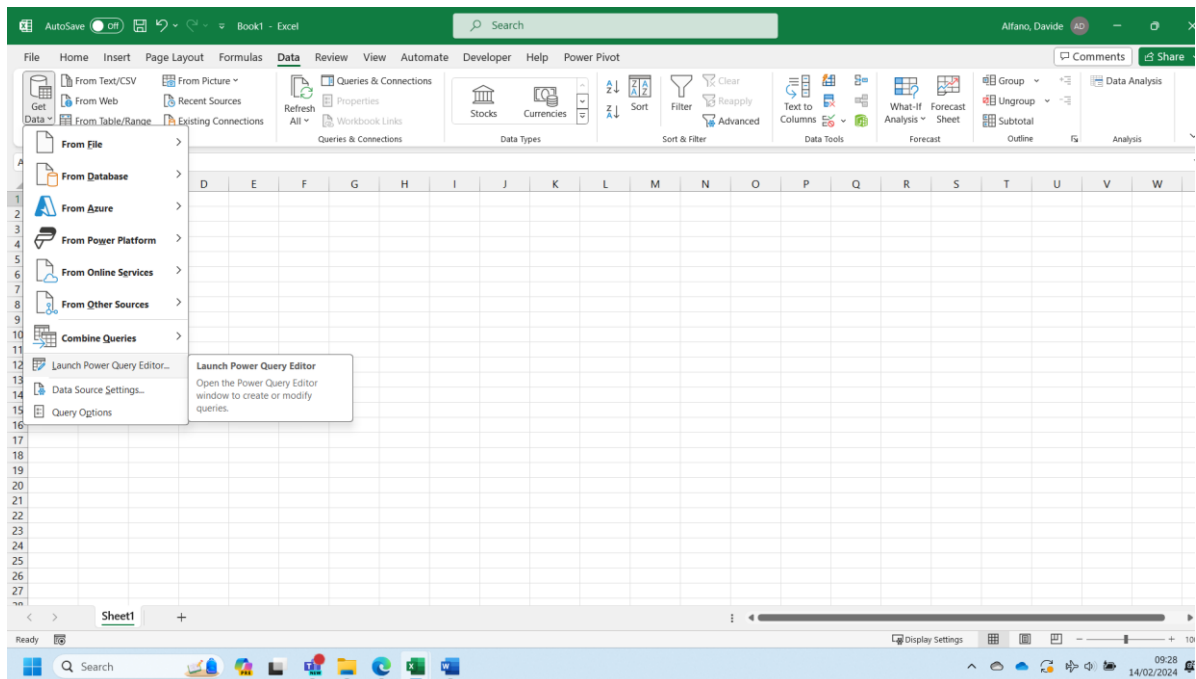


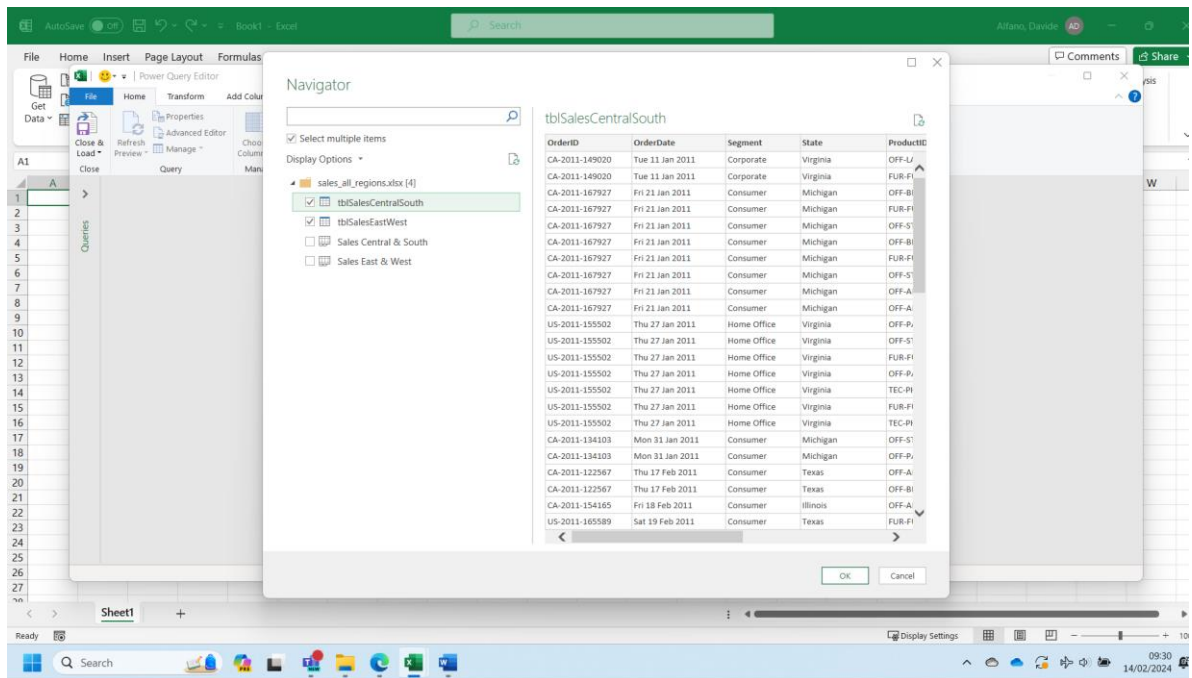
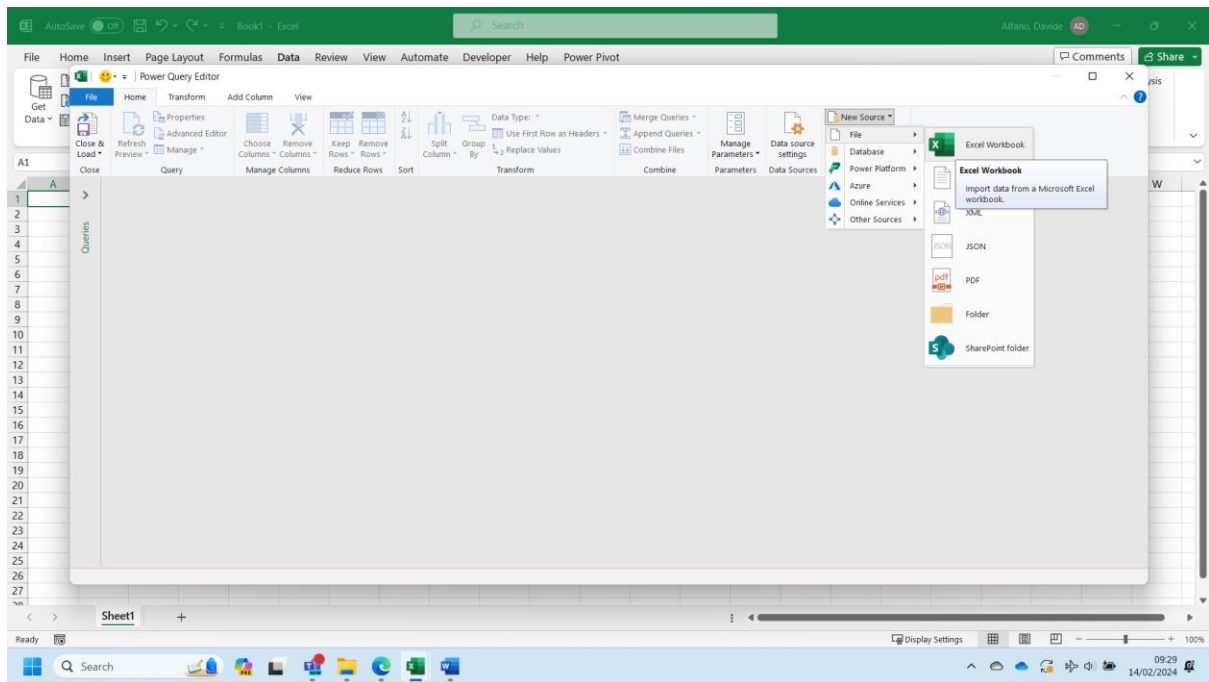
Activity—Scenario Demonstration Practice – Day 1

Part 1 – Gather data and create the Sales table

In the 'data' folder of provided files, locate the Excel workbook 'sales_all_regions.xlsx'. The workbook contains two tables: tblSalesEastWest and tblSalesCentralSouth.

- Import both tables into Power Query Editor. Name them 'SalesEW' and 'SalesCS'.





- How many rows are in SalesEW and in SalesCS?

Use Count Rows under the tab Transform in Power Query Editor to get the information.

Power Query Editor interface showing a table with columns: OrderID, OrderDate, State, ProductID, Quantity, and DiscountPer. The table contains 999 rows of data. The interface includes a ribbon with tabs like File, Home, Insert, Page Layout, Formulas, Data, Review, View, Automate, Developer, Help, and Power Pivot. The Data tab is active, showing various transformation options. A 'Count Rows' query is selected, and the 'Query Settings' pane on the right shows the 'Changed Type' step.

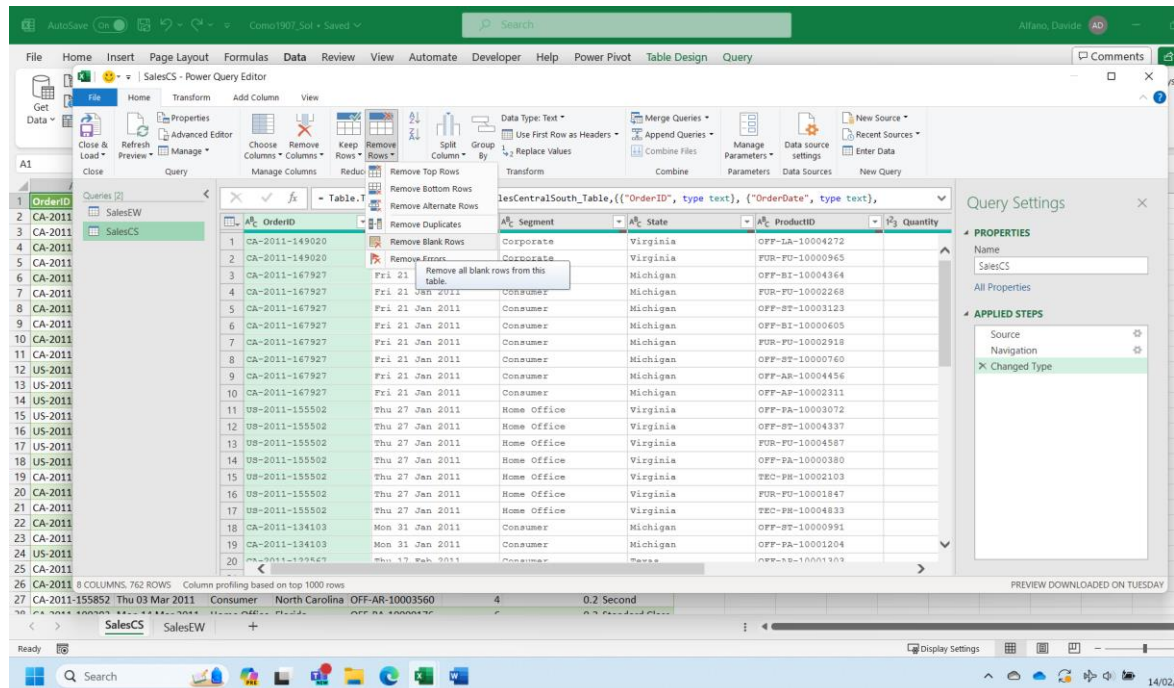
OrderID	OrderDate	State	ProductID	Quantity	DiscountPer
CA-2011-115791	Mon 17 Jan 2011	Pennsylvania	FUR-FU-10001095	6	
CA-2011-115791	Mon 17 Jan 2011	Pennsylvania	TEC-PH-10004614	3	
CA-2011-115791	Mon 17 Jan 2011	Pennsylvania	OFF-LA-10001074	3	
CA-2011-115791	Mon 17 Jan 2011	Pennsylvania	OFF-BI-10001575	2	
CA-2011-104808	Sat 05 Feb 2011	California	OFF-BI-10003676	2	
CA-2011-107181	Sat 05 Feb 2011	California	OFF-PA-10000350	4	
CA-2011-107181	Sat 05 Feb 2011	California	OFF-BI-10004230	3	
CA-2011-133354	Wed 23 Feb 2011	California	OFF-PA-10001800	3	
US-2011-137690	Fri 25 Feb 2011	Oregon	OFF-PA-10000069	3	
US-2011-137690	Fri 25 Feb 2011	Oregon	OFF-PA-10000174	4	
CA-2011-104269	Tue 01 Mar 2011	Washington	FUR-CH-10004063	2	
US-2011-127978	Thu 03 Mar 2011	Ohio	OFF-LA-10000305	3	
US-2011-127978	Thu 03 Mar 2011	Ohio	FUR-BO-10001972	5	
US-2011-127978	Thu 03 Mar 2011	Ohio	OFF-ST-10002486	8	
CA-2011-104563	Mon 07 Mar 2011	Washington	OFF-ST-10000934	5	
CA-2011-104563	Mon 07 Mar 2011	Washington	FUR-CH-10002780	6	
CA-2011-104563	Mon 07 Mar 2011	Washington	OFF-AR-10000390	5	

SalesEW: 1168

SalesCS: 762

- Remove the blank rows from SalesEW, SalesCS. How many blank rows are removed?

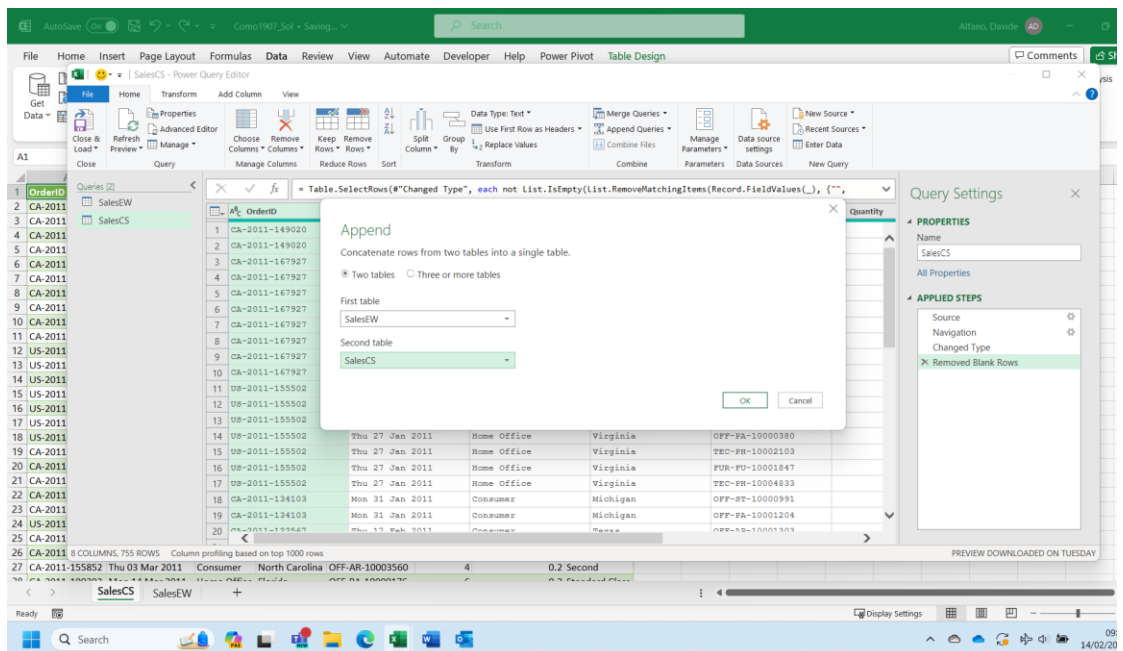
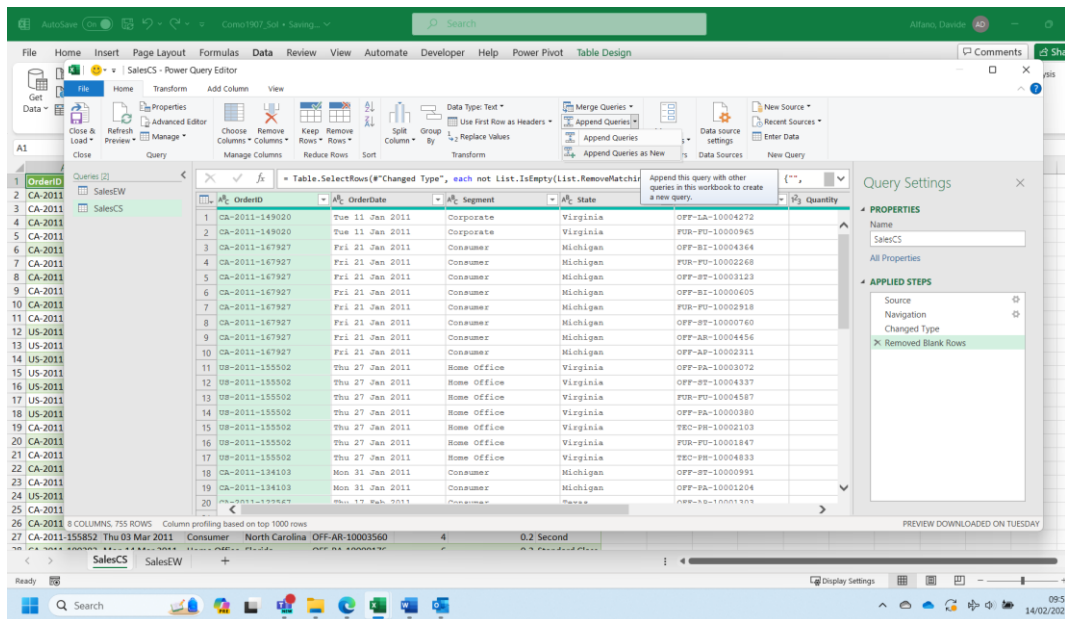
Use Remove Blank Rows under Transform/Remove Rows to get the information required.



SalesEW: 25 (initial number of rows: 1168 - current number of rows: 1143)
SalesCS: 7 (762-755)

- Next, we need to append SalesCS to SalesEW. This means that we add the rows of SalesCS to the bottom of SalesEW.

Let's use Append Queries as New in Power Query Editor. The difference with Append Queries is that a new table is created because of it, without any change to the existing tables.



- Which column of SalesCS does not match with any column of SalesEW?

OrderID	OrderDate	State	ProductID	Quantity	DiscountPercent	ShipMode	Segment
281	Mon 09 Apr 2012	Connecticut	OFF-PA-1000179	3		0 Standard	
282	US-2012-136987	Wed 11 Apr 2012	California	OFF-PA-10001127	2	0 Second Class	
283	US-2012-136987	Weds 11 Apr 2012	California	TEC-CO-10001943	4	0.2 Standard Class	
284	CA-2012-133837	Fri 13 Apr 2012	Arizona	OFF-ST-10000344	1	0.2 Standard	
285	US-2012-120502	Fri 13 Apr 2012	California	TEC-PH-10003645	2	0.2 Standard Class	
286	US-2012-120502	Fri 13 Apr 2012	California	OFF-AP-10004980	2	0 Standard Class	
287	US-2012-120502	Fri 13 Apr 2012	California	FUR-RU-10004973	2	0 Standard Class	
288	CA-2012-161263	Mon 16 Apr 2012	Ohio	OFF-BI-10004330	5	0.7 Standard Class	
289	CA-2012-161263	Mon 16 Apr 2012	Ohio	OFF-FA-10004838	7	0.2 Standard	
290	CA-2012-161263	Mon 16 Apr 2012	Ohio	TEC-PH-10002115	3	0.4 Standard Class	
291	CA-2012-161263	Mon 16 Apr 2012	Ohio	OFF-FA-10003059	1	0.2 Standard Class	
292	CA-2012-161263	Mon 16 Apr 2012	Ohio	OFF-AR-10004757	3	0.2 Standard	
293	CA-2012-161263	Mon 16 Apr 2012	Ohio	OFF-AP-10002350	3	0.2 Standard	
294	CA-2012-148712	Tue 17 Apr 2012	New York	TEC-AC-10003198	1	0 Standard Class	
295	CA-2012-148712	Tue 17 Apr 2012	New York	OFF-AR-10000614	3	0 Standard Class	
296	CA-2012-148712	Tue 17 Apr 2012	New York	OFF-BI-10001900	13	0.2 Standard	
297	CA-2012-130716	Mon 30 Apr 2012	California	TEC-PH-10000347	5	0.2 Standard Class	
298	CA-2012-130716	Mon 30 Apr 2012	California	OFF-AR-10000658	3	0 Standard Class	
299	CA-2012-144043	Thu 10 May 2012	Colorado	TEC-AC-10002718	4	0.2 Standard Class	
300	CA-2012-130876	Sat 12 May 2012	New York	OFF-ST-10004258	3	0 Second Class	
301	CA-2012-162607	Sat 12 May 2012	Washington	OFF-BI-10002949	3	0.2 Standard Class	
302	CA-2012-119291	Mon 14 May 2012	Pennsylvania	OFF-LA-10003510	4	0.2 First	
303	CA-2012-119291	Mon 14 May 2012	Pennsylvania	OFF-AR-10001118	1	0.2 First	
304	CA-2012-119291	Mon 14 May 2012	Pennsylvania	OFF-BI-10001575	2	0.7 First Class	
305	CA-2012-119291	Mon 14 May 2012	Pennsylvania	OFF-BI-10001982	3	0.7 First	
306	---	Mon 14 May 2012	Pennsylvania	OFF-AR-10003373	8	0.2 First	
307	CA-2012-119291	Mon 14 May 2012	Pennsylvania	OFF-EN-10000927	7	0.2 First	

The only unmatched column between the two tables is segment. Delete it from Sales.

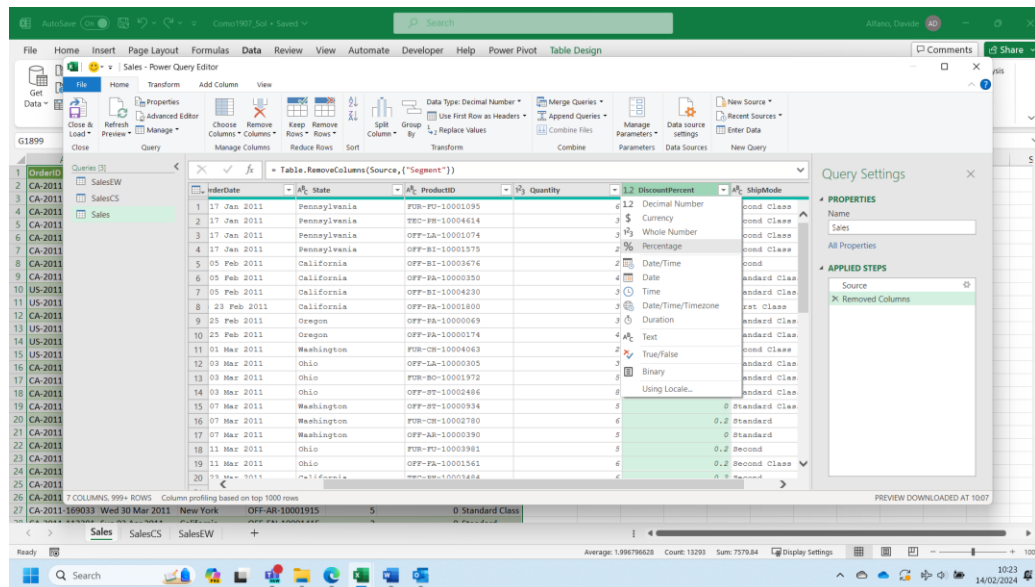
- Verify that Sales contains 7 columns and 1,898 rows, and that all columns were appended in the correct positions.

OrderID	OrderDate	State	ProductID	Quantity	DiscountPercent	ShipMode
CA-2012						
CA-2011						
CA-2011						
CA-2011						
CA-2011						
CA-2011						
CA-2011						
CA-2011						
CA-2011						
US-2011						
US-2011						

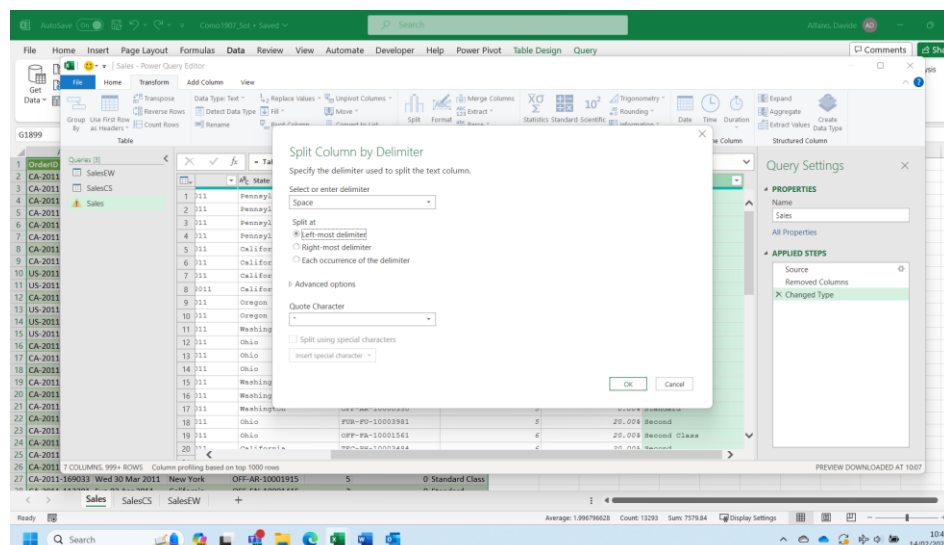
Part 2 – Clean the Sales table

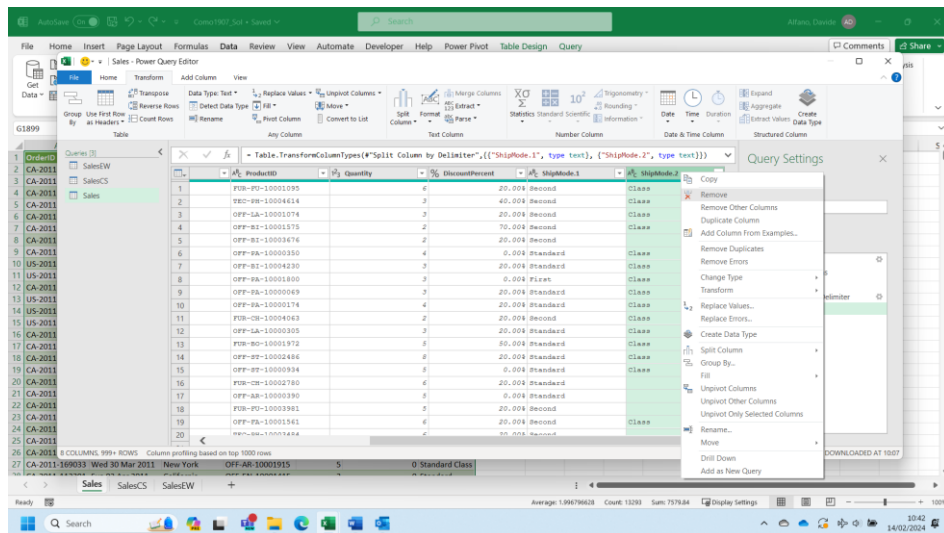
- Inspect the Sales table. Ensure that the [Quantity] column is formatted as an integer (whole number). Ensure that the [DiscountPercent] column has been correctly parsed as a numeric percentage (not just a string with '%' at the end).

We can easily modify the data type in Power Query Editor selecting the appropriate option on the left top corner of each column header.

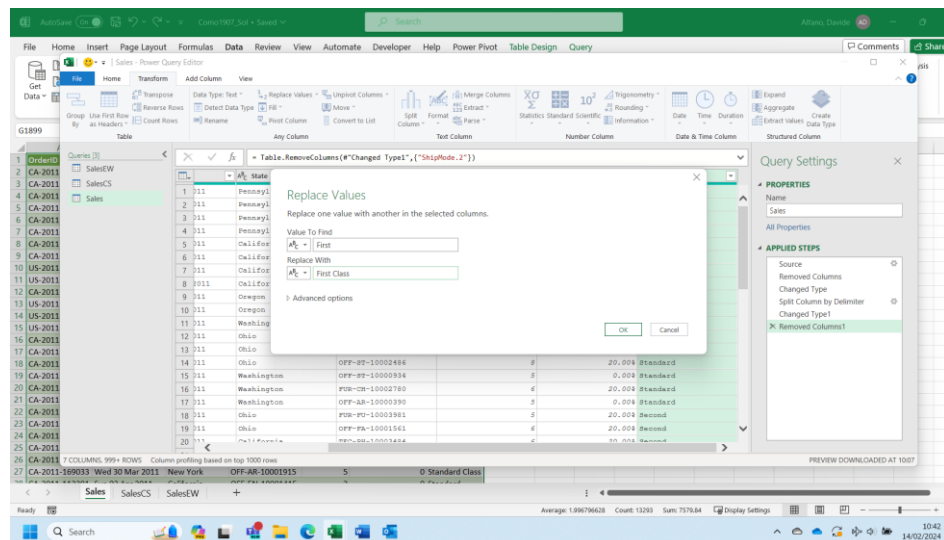


- In the [ShipMode] column, the values 'First', 'Second' and 'Standard' should instead be 'First Class', 'Second Class' and 'Standard Class'. Make this correction. Use Replace Values in Power Query Editor. You will probably need to split the [ShipMode] column in two columns before, eliminating the word Class, where available, from the first column. The second column originating from the split will be dropped.

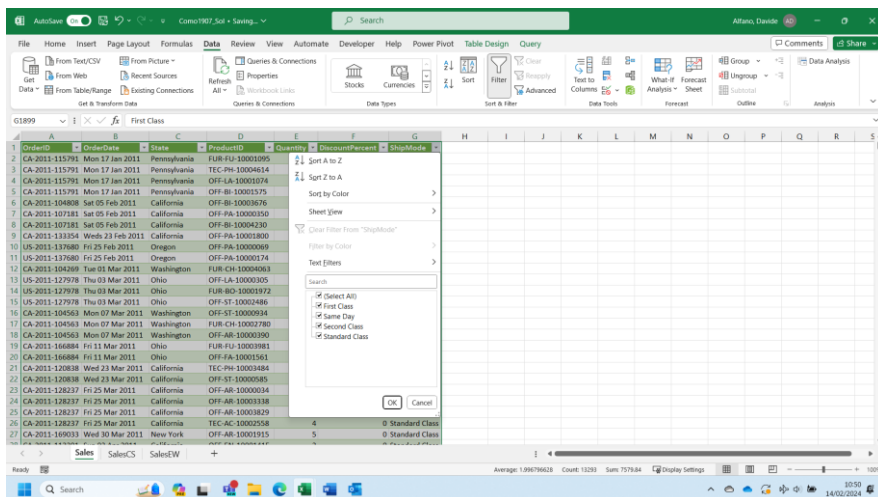




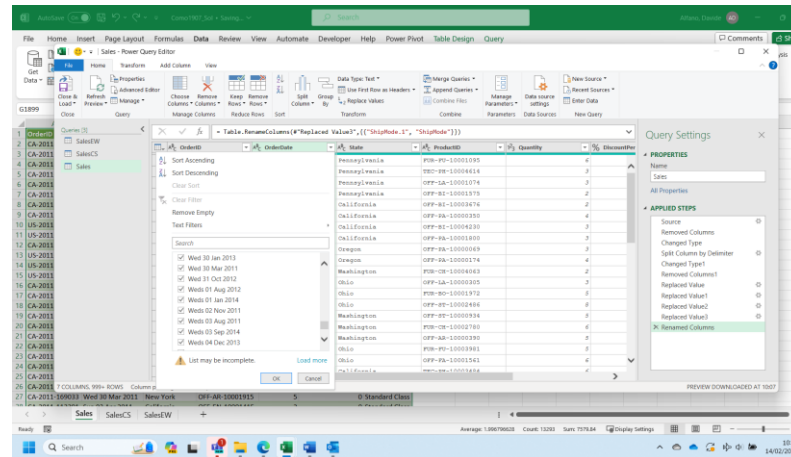
Now we can use Replace Values to replace First Class to First, Second Class to Class ...



Because of the correction, the only distinct values in this column are now 'First Class', 'Second Class', 'Standard Class' and 'Same Day'.



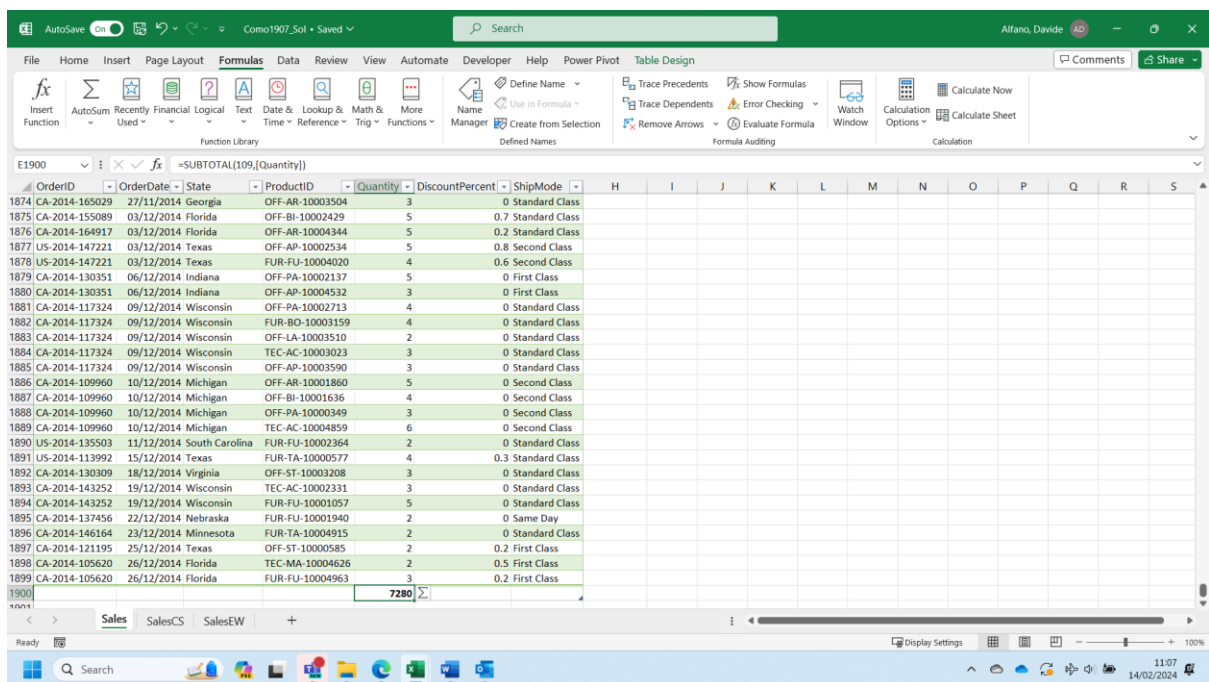
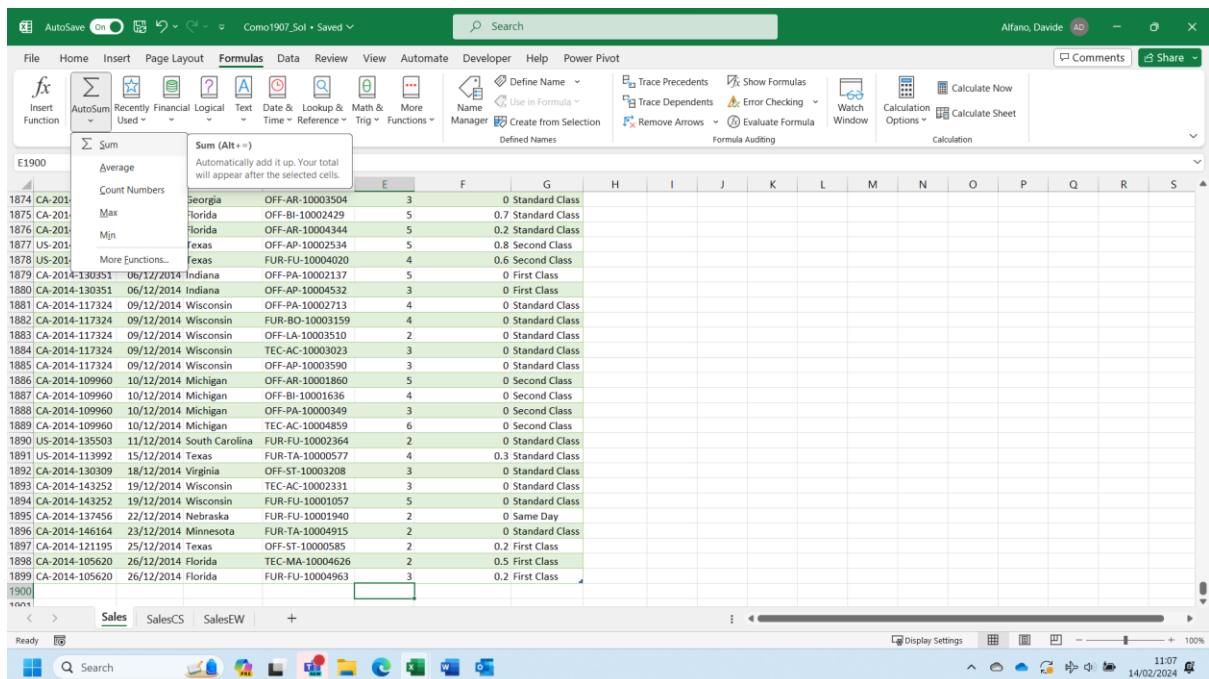
- The [OrderDate] column contains dates formatted as text. These values need to be converted to machine-readable dates. Perform this conversion. If not all [OrderDate] values were successfully converted to dates, explain what the problem was. Fix the problem.



We can easily modify the data type in Power Query Editor, by selecting the appropriate option (Date) on the left top corner of the column header. However, some records are improperly using the abbreviation 'Weds' for Wednesday, instead of 'Wed'. 'Weds' is not treated as a valid abbreviation for Wednesday by Power Query Editor when implementing the conversion of the field to Date. We must fix this replacing Weds with Wed using Replace Values. Afterwards, we can change the datatype as previously described.

Part 3 – Analysis

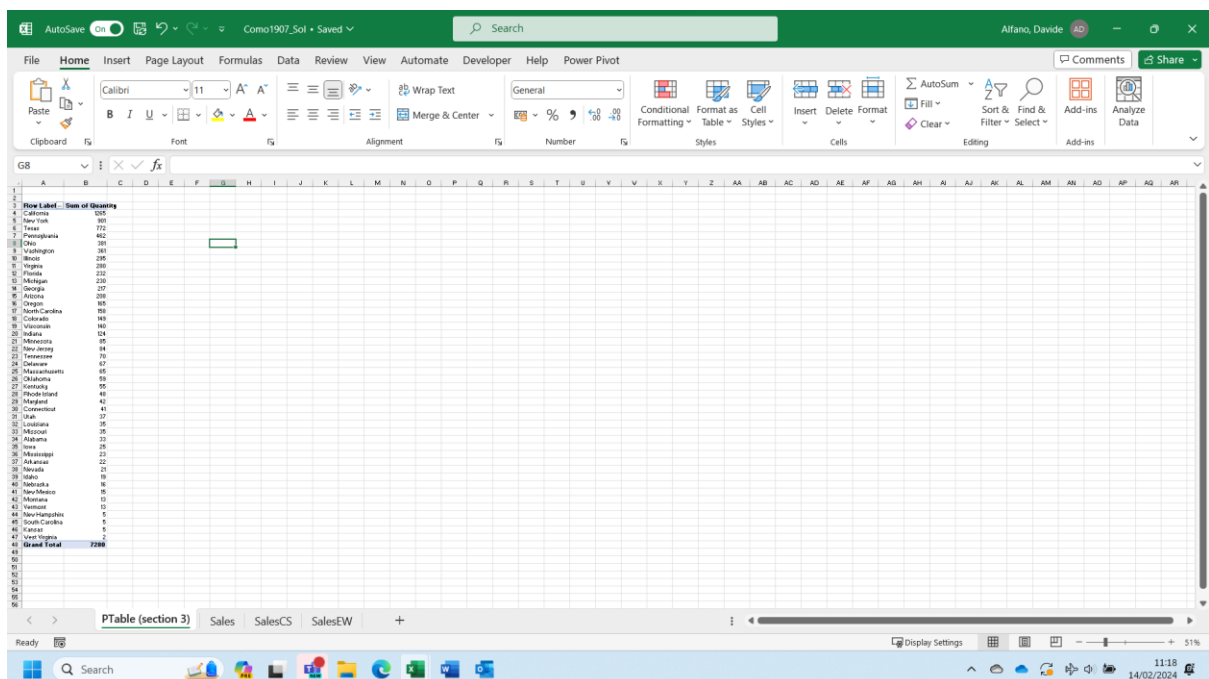
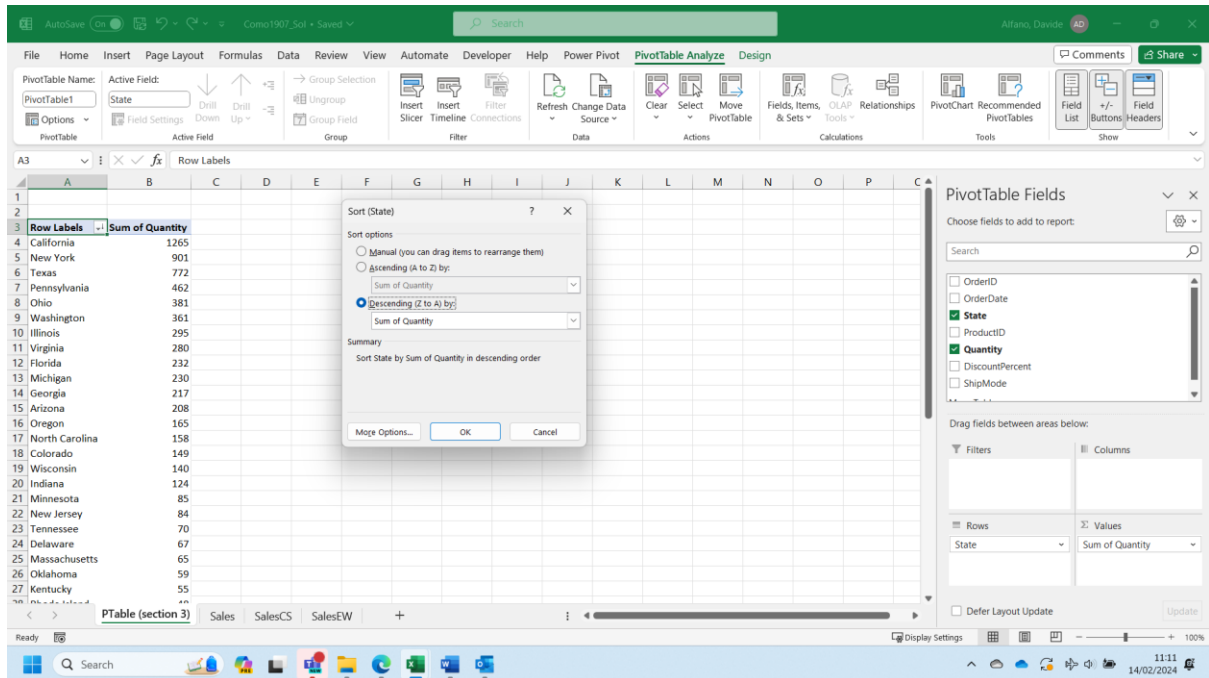
- How many units were sold in total (sum of [Quantity])?



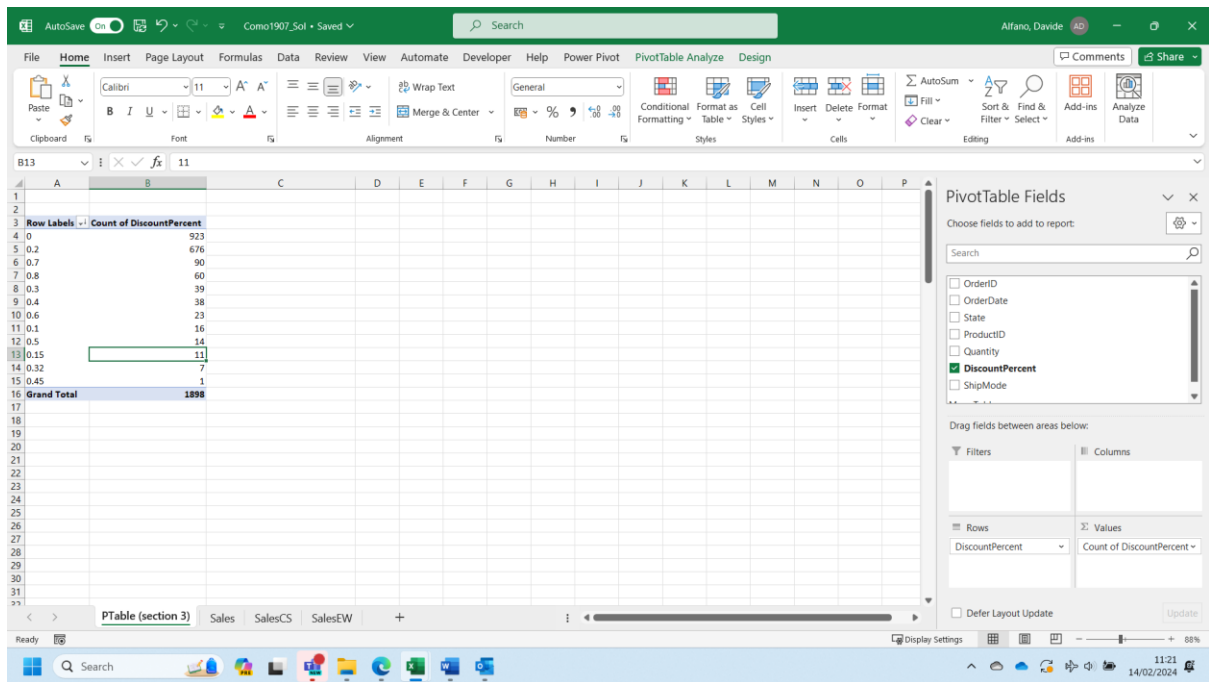
The AutoSum function (under the tab Formulas in Excel) gives us the result of 7280 units sold in total.

- Which state has purchased the most units? Which state has purchased the fewest units?

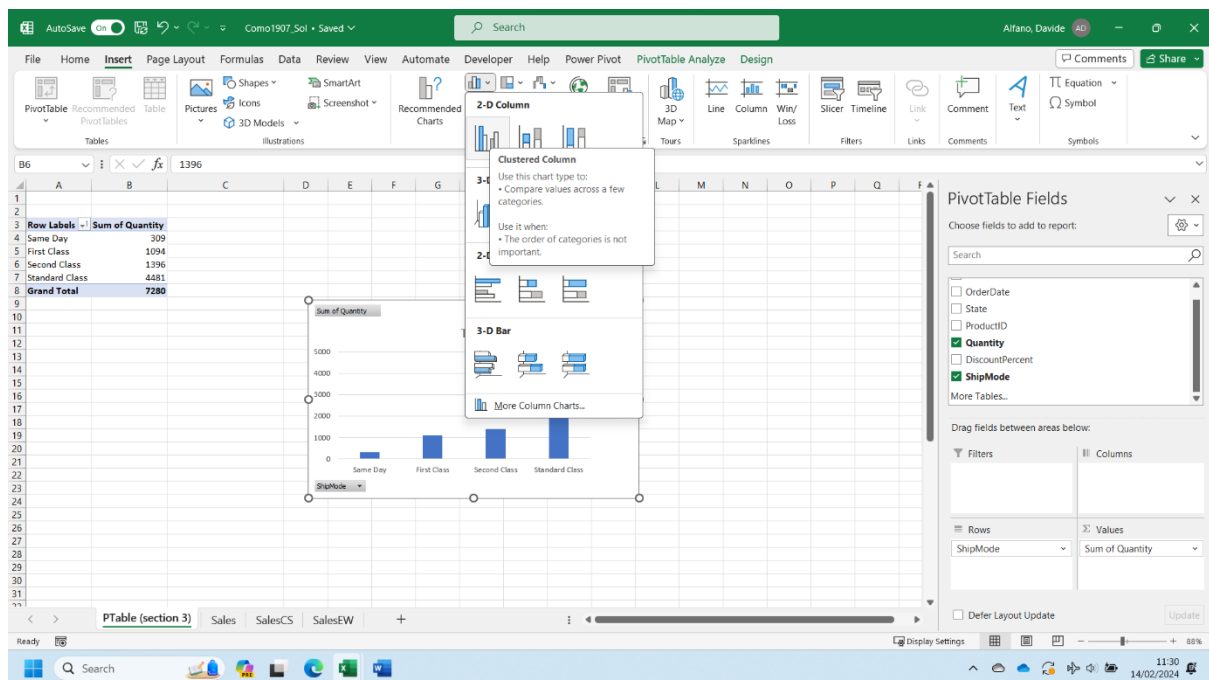
A Pivot Tables can help you in determining the total number of units sold in each state ('sum of quantity'). Values will have to be re-ordered according to the 'sum of quantity'.



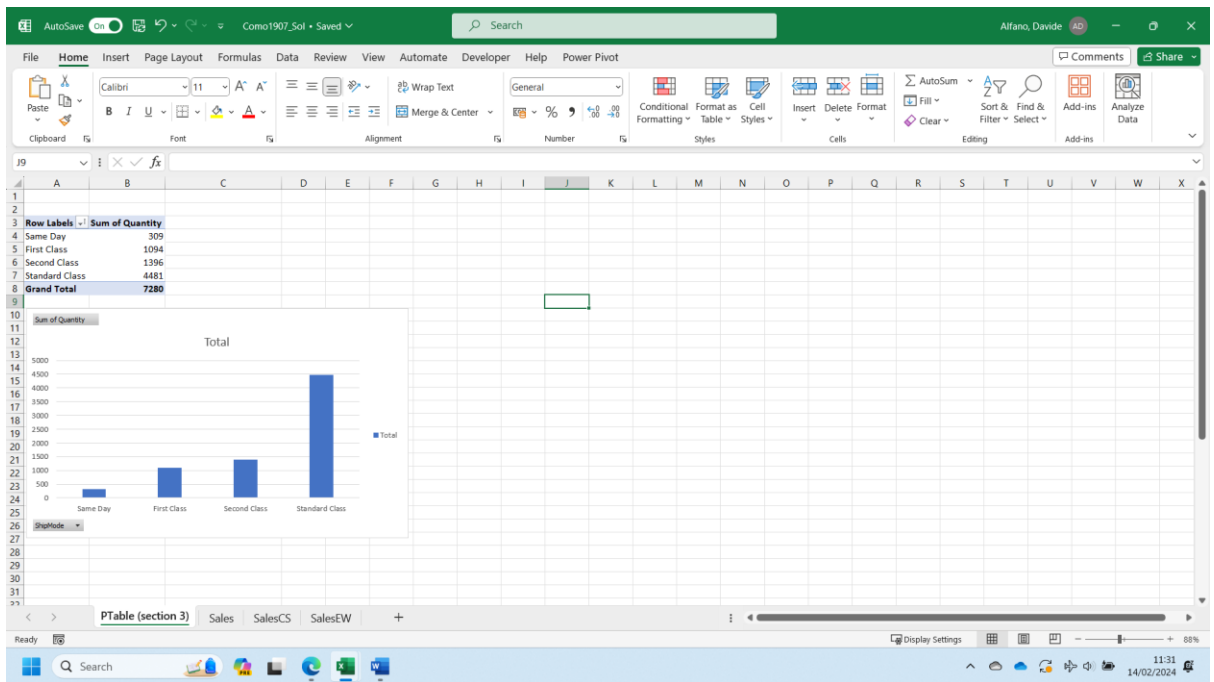
- What are the three most common [DiscountPercent] values?
- Once again, the best option is to use a Pivot Table as defined below:



- How many units were shipped by each [ShipMode]? Plot a suitable chart. The Pivot Table that has been created will have to be re-designed as shown below. Based on the same Table, a Graph will be added to visualize the

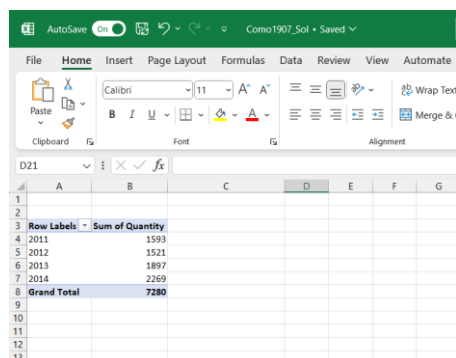
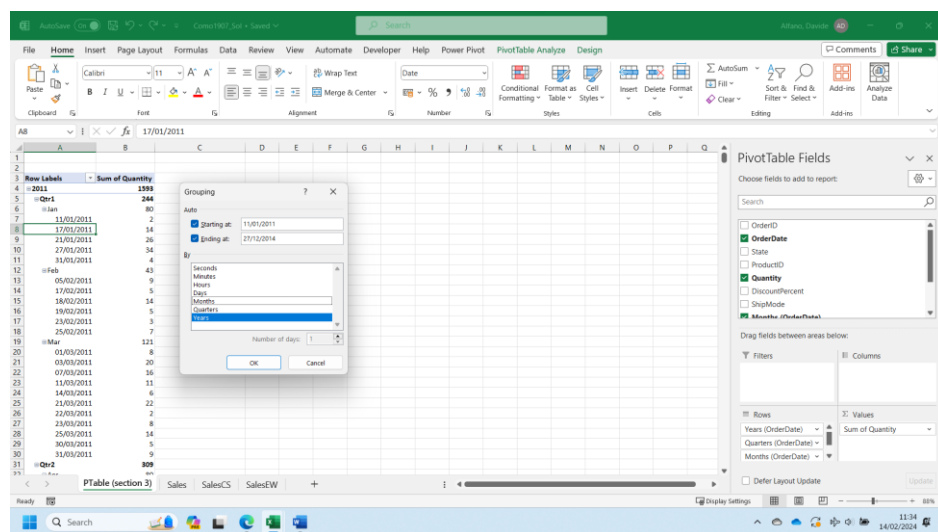


information required.



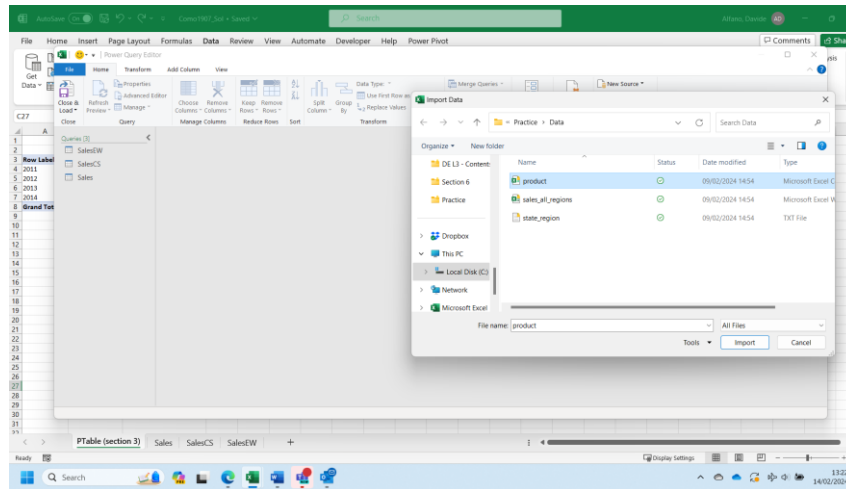
- How many units were shipped each year? Plot a suitable chart. Briefly describe the trend over time from 2011 to 2014.

PivotTables can perform date aggregation by Year, Month, Day etc. Right-click somewhere inside a date column of a PivotTable and select Group....

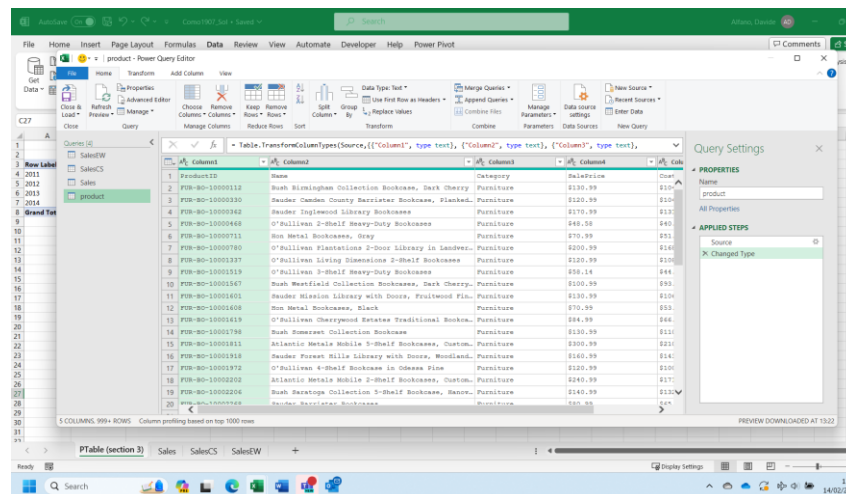


Part 4 – Clean the Product Table

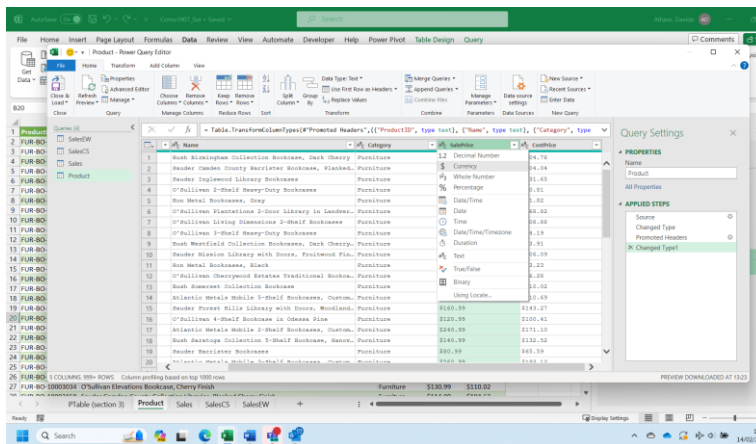
- Import this file. Name the table 'Product'.



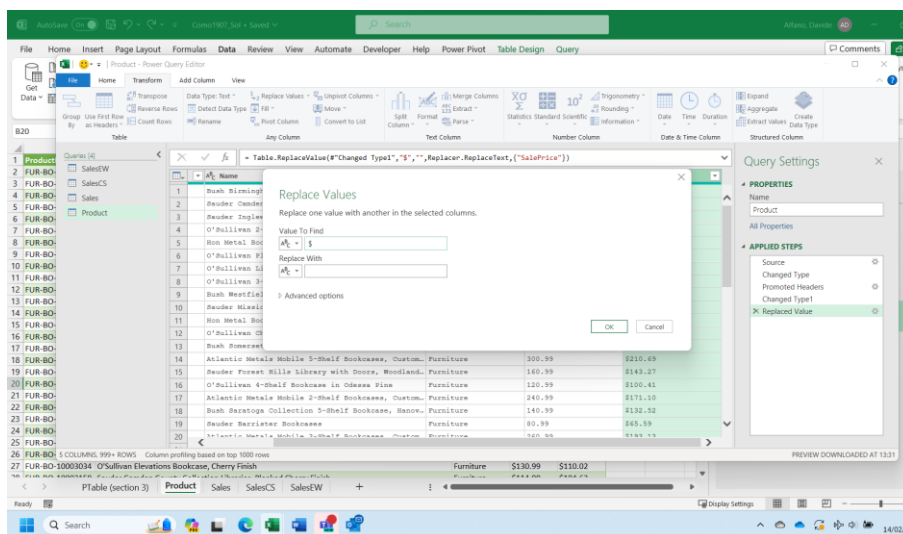
The first row will have to be promoted as header (Use First Row as Header under the Tab Home)



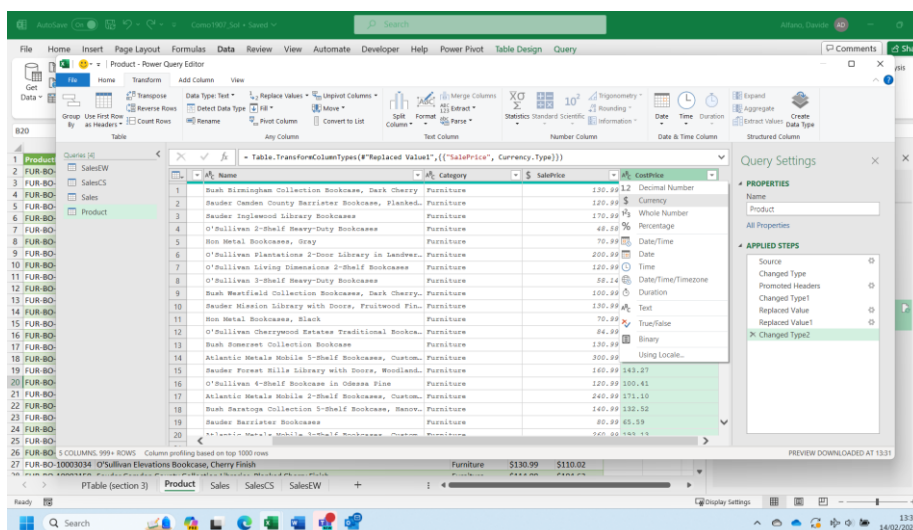
- The [SalePrice] and [CostPrice] columns need to be converted to a numeric type (Currency is a suitable type). Perform this conversion.



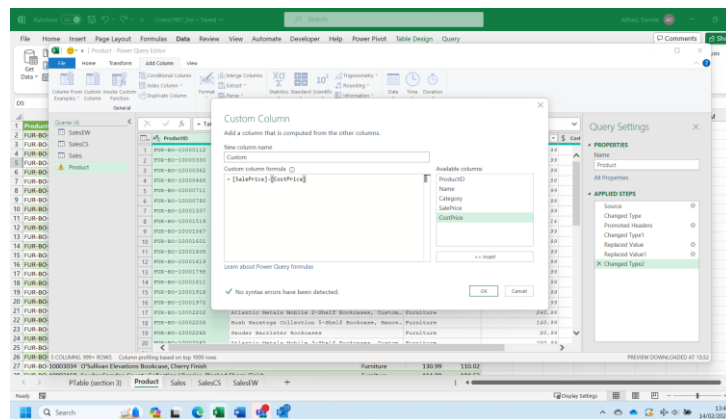
Literal '\$' character prevented conversion to a numeric type. Solution: replace '\$' with ',', i.e. remove the '\$' symbol.



After this change the conversion to currency can be performed.



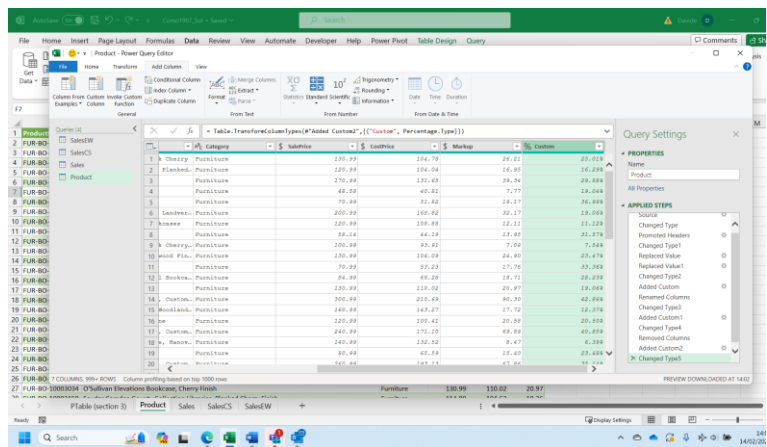
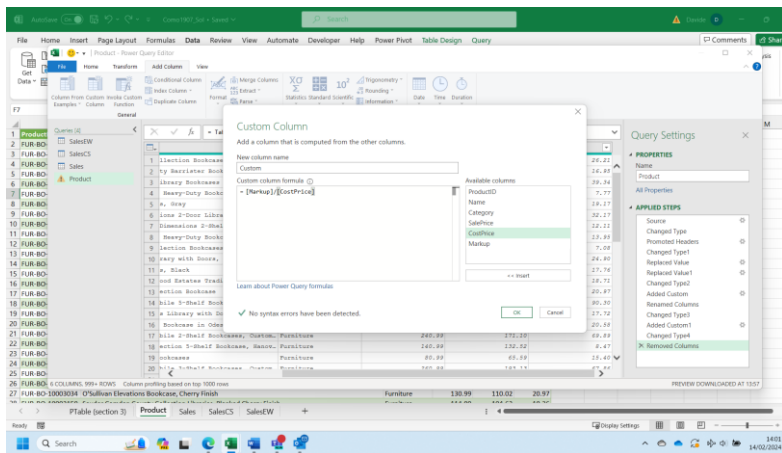
- Create a new column, [Markup], calculated as [SalePrice] – [CostPrice]. Round this column to 2 decimal places to eliminate minor rounding errors by converting it to an appropriate type.



Product	Category	Subcategory	SalePrice	CostPrice	Markup
1. Adventure Works Books, Dark Cherry	Paperback	Paperback	120.99	204.70	24.21
2. 192 Recliner Bookcase, Elmwood	Paperback	Paperback	120.99	204.70	14.49
3. Library Bookcase	Paperback	Paperback	170.99	130.49	34.24
4. Heavy-Duty Bookcase	Paperback	Paperback	40.99	40.49	5.77
5. A. Gray	Paperback	Paperback	70.99	51.82	14.17
6. 1000 2-Door Library in Laminate	Paperback	Paperback	200.99	144.42	32.17
7. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	22.12
8. Heavy-Duty Bookcase	Paperback	Paperback	58.14	44.18	13.99
9. Section Bookcase, Dark Cherry	Paperback	Paperback	120.99	99.42	7.49
10. Gray with Stone, Preshield Plus	Paperback	Paperback	120.99	104.99	24.90
11. A. Black	Paperback	Paperback	70.99	63.23	17.74
12. 1000 2-Door Library in Laminate	Paperback	Paperback	44.99	44.49	24.71
13. Section Bookcase	Paperback	Paperback	120.99	100.99	20.97
14. 1000 2-Door Library in Laminate	Paperback	Paperback	200.99	200.49	40.20
15. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	144.42	27.72
16. Bookcase in Glass and Steel	Paperback	Paperback	120.99	100.42	20.99
17. 1000 2-Door Library in Laminate	Paperback	Paperback	240.99	170.10	69.49
18. 1000 2-Door Library in Laminate	Paperback	Paperback	240.99	200.99	4.47
19. Bookcase	Paperback	Paperback	80.99	60.49	15.49
20. 1000 2-Door Library in Laminate	Paperback	Paperback	200.99	170.10	15.49
21. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
22. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
23. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
24. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
25. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
26. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
27. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
28. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
29. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
30. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97
31. 1000 2-Door Library in Laminate	Paperback	Paperback	120.99	100.99	20.97

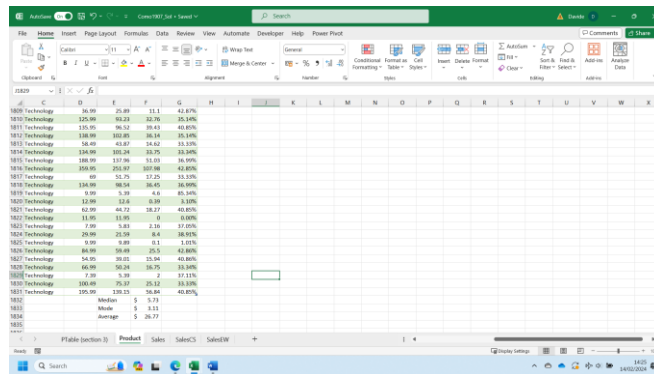
- Create a new column, [MarkupPercent], which expresses [Markup] as a fraction of [CostPrice].

A [MarkupPercent] of 0 (or 0%) means that [SalePrice] is equal to [CostPrice]. A [MarkupPercent] of 1 (or 100%) means that [SalePrice] is double [CostPrice]. Convert [MarkupPercent] to a suitable numeric type. It is not necessary to round this column.



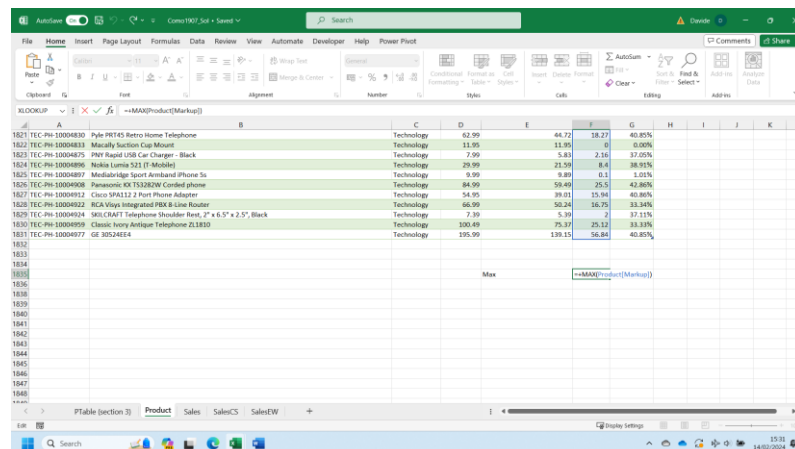
Part 5 – Analysis

- What are the mean, median and mode of [Markup] (in \$)?
Use the Excel formulas median(), MODE.SNGL(), average() to estimate those values



Product	Price	Cost	Markup
1809 Technology	86.99	25.89	11.1
1810 Technology	325.99	93.23	32.76
1811 Technology	175.99	96.52	19.43
1812 Technology	148.99	103.85	16.14
1813 Technology	154.99	43.67	14.52
1814 Technology	134.99	105.24	13.75
1815 Technology	188.99	117.96	15.53
1816 Technology	355.95	251.87	107.08
1817 Technology	89	52.75	17.25
1818 Technology	134.99	98.54	36.45
1819 Technology	9.99	5.39	4.6
1820 Technology	12.99	12.4	0.59
1821 Technology	42.99	42.17	28.57
1822 Technology	11.95	11.95	0
1823 Technology	9.99	5.89	4.1
1824 Technology	29.99	25.59	8.4
1825 Technology	54.99	30.01	23.94
1826 Technology	84.99	56.49	25.5
1827 Technology	66.99	50.24	16.75
1828 Technology	66.99	50.24	16.75
1829 Technology	100.49	75.17	25.32
1830 Technology	139.15	106.24	42.85%
1831	Median		5.73
1832	Mode		5.11
1833	Average		5.26.77

- Which product has the highest [Markup]? There are no ties. Apply appropriate number formatting.
The first step requires to determine the highest Markup value. You can use the Excel function Max().



Product	Price	Cost	Markup
1821 TEC-PH-10004810 Pyle PR145 Retro Home Telephone	Technology	42.99	44.71
1822 TEC-PH-10004813 Musically Section Cap Mount	Technology	11.95	11.95
1823 TEC-PH-10004817 Pyle Rapid USB Car Charger - Black	Technology	7.99	5.83
1824 TEC-PH-10004896 Audio Linea 3.21 (1 Model)	Technology	29.99	21.56
1825 TEC-PH-10004897 Modularidge Sport Armband iPhone 5s	Technology	9.99	9.89
1826 TEC-PH-10004909 Panasonic DECT2.0GHz Corded phone	Technology	84.99	59.49
1827 TEC-PH-10004912 Cisco SPA112 2 Port Phone Adapter	Technology	54.95	39.01
1828 TEC-PH-10004922 RCA Viqas Integrated PRX.6 Line Router	Technology	66.99	50.24
1829 TEC-PH-10004924 160CMx47" Telephone Shoulder Rest, 17" x 6.5" x 2.5", Black	Technology	7.99	5.39
1830 TEC-PH-10004999 Classic luxury Antique Telephone ZL1830	Technology	100.49	75.17
1831 TEC-PH-10004977 GE 30524614	Technology	139.15	106.24
1832			
1833			
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Secondly you will have to find the corresponding product using the Xlookup () function.

Excel interface showing a table of products. The formula bar displays: `=MAX(UPPER(Sales[ProductID],ProductID[ProductID]))`. The table includes columns for ProductID, Product, Price, Markup, and MarkupPercent. The bottom of the table shows a summary row with 'Max' and a value of 33.34%.

- Which product has the highest [MarkupPercent]? (There are no ties.)

The steps to implement are the same as for the previous question. You only need to change the column of reference.

Excel interface showing a filtered view of the product table. The formula bar displays: `=MAX(UPPER(Sales[ProductID],ProductID[ProductID]))`. The table includes columns for ProductID, Product, Price, Markup, and MarkupPercent. The bottom of the table shows a summary row with 'Max' and a value of 100.50%.

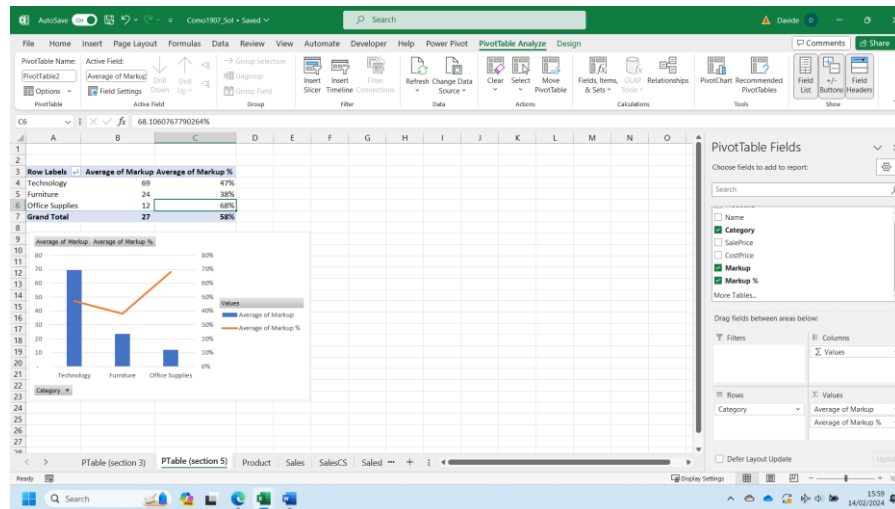
- What is the lowest [Markup], and how many products share it?
Use the Excel function min() to determine the lowest Markup, and the function countif() to determine the number of products having the lowest Markup.

Excel interface showing a filtered view of the product table. The formula bar displays: `=COUNTIF(Product[Markup],0)`. The table includes columns for ProductID, Product, Price, Markup, and MarkupPercent. The bottom of the table shows a summary row with 'Min' and a value of 0.

- Find the mean [Markup] and mean [MarkupPercent] by [Category]. Plot one or more suitable charts.

Which [Category] has the highest mean [Markup]? Which [Category] has the highest mean [MarkupPercent]?

A Pivot Table can help you in answering such a question in straightforward way.



Activity—Scenario Demonstration Practice – Day 2

Part 6 – Merge Sales to Product

- Create a new table by merging the Sales table to the Product table on an appropriate join column. Use a left join, with Sales as the left table. Name the new table 'SalesDetails'.

Table.TransformColumns(Sales, {"ProductID", type text})

OrderID	OrderDate	State	ProductID	Quantity	DiscountPercent
CA-2011-115791	27/02/2012	Pennsylvania	PID-PD-10001395	4	
CA-2011-115791	27/02/2012	Pennsylvania	PID-PD-10004614	3	
CA-2011-115791	27/02/2012	Pennsylvania	OPF-SA-10001774	2	
CA-2011-115791	27/02/2012	Pennsylvania	OPF-BI-10001370	2	
CA-2011-104808	08/02/2012	California	OPF-BI-10001876	2	
CA-2011-107181	08/02/2012	California	OPF-PA-10000200	4	
CA-2011-107181	08/02/2012	California	OPF-BI-10000330	3	
CA-2011-113334	23/02/2012	California	OPF-PA-10001800	3	
OP-2011-117480	25/02/2012	Oregon	OPF-PA-10000049	3	
OP-2011-117480	25/02/2012	Oregon	OPF-PA-10001174	4	
OP-2011-104249	01/03/2012	Washington	PID-OR-10004063	2	
OP-2011-127978	03/03/2012	Oahu	OPF-SA-10000305	3	
OP-2011-127978	03/03/2012	Oahu	PID-BO-10001972	3	
OP-2011-127978	03/03/2012	Oahu	OPF-BI-10000466	4	
OP-2011-104543	07/03/2012	Washington	OPF-BI-10000924	3	
OP-2011-104543	07/03/2012	Washington	PID-BO-10002780	4	
CA-2011-104543	07/03/2012	Washington	OPF-AP-10000290	3	
CA-2011-166884	11/03/2012	Oahu	PID-PO-10001981	3	
CA-2011-166884	11/03/2012	Oahu	OPF-PA-10001361	3	
CA-2011-166884	11/03/2012	Oahu	OPF-PA-10001361	3	

Select tables and matching columns to create a merged table.

Tables: Sales, Product

Join Kind: Left Outer (all from first, matching from second)

Use Fuzzy matching to perform the merge

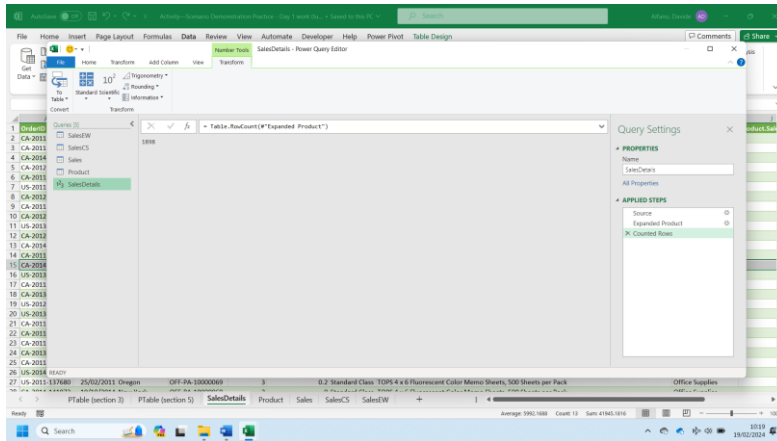
Fuzzy matching options

The selection matches 1000 of 1000 rows from the first table.

Table.NestedJoin(Sales, ("ProductID"), Product, ("ProductID"), "Product", JoinKind.LeftOuter)

ProductID	Quantity	DiscountPercent	ShipDate	Product
PID-PO-10001395	4	20.00%	Second Class	Table
PID-PA-10004614	3	40.00%	Second Class	Table
OPF-SA-10001774	2	20.00%	Second Class	Table
OPF-BI-10001370	2	20.00%	Second Class	Table
OPF-PA-10000200	4	0.00%	Second Class	Table
OPF-BI-10000330	3	20.00%	Second Class	Table
OPF-PA-10001800	3	20.00%	Second Class	Table
OPF-PA-10000049	3	20.00%	Second Class	Table
OPF-PA-10001174	4	20.00%	Second Class	Table
PID-OR-10004063	2	20.00%	Second Class	Table
OPF-SA-10000305	3	20.00%	Second Class	Table
PID-BO-10001972	3	20.00%	Second Class	Table
OPF-BI-10000466	4	20.00%	Second Class	Table
OPF-BI-10000924	3	20.00%	Second Class	Table
PID-BO-10002780	4	20.00%	Second Class	Table
OPF-AP-10000290	3	20.00%	Second Class	Table
PID-PO-10001981	3	20.00%	Second Class	Table
OPF-PA-10001361	3	20.00%	Second Class	Table
OPF-PA-10001361	3	20.00%	Second Class	Table

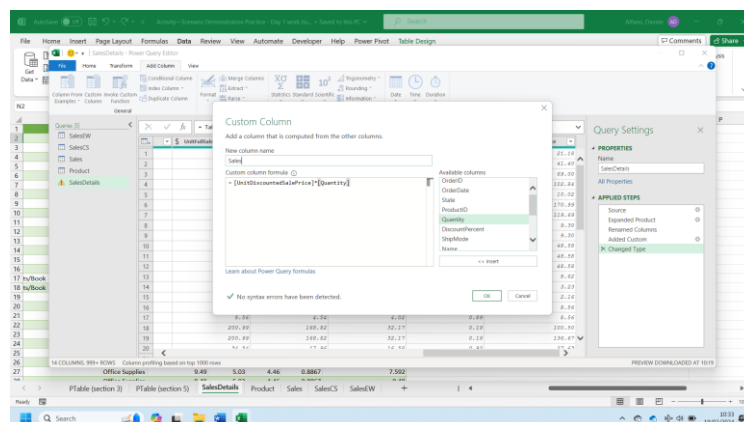
Total number of columns /rows: 13/1898



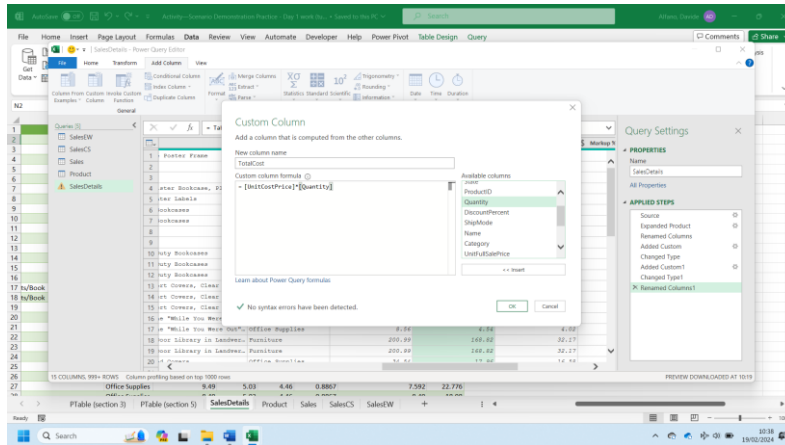
- In the SalesDetails table, rename the [SalePrice] column 'UnitFullSalePrice'. This is the undiscounted sale price of a single unit of the product.
- Create a new column, [UnitDiscountedSalePrice], and populate it with the discounted sale price of a single unit of the product. Round this column to 2 decimal places to prevent invalid unit prices.

Category	UnitFullSalePrice	CostPrice	Markup	Markup %	UnitDiscountedSalePrice	Quantity
Furniture	26.48	16.42	10.06	0.8127	25.184	69
Technology	69	51.75	17.25	0.3333	41.4	69
Furniture	120.99	104.04	16.95	0.1629	102.8415	120.99
Office Supplies	12.53	6.64	5.89	0.887	10.024	170.99
Furniture	170.99	111.65	59.34	0.347	110.99	30.99
Office Supplies	30.99	16.11	14.88	0.9236	9.297	30.99
Office Supplies	30.99	16.11	14.88	0.9236	9.297	48.58
Furniture	48.58	40.81	7.77	0.1994	48.58	10.78
Office Supplies	10.78	5.61	5.17	0.9216	3.234	10.78
Office Supplies	10.78	5.61	5.17	0.9216	3.234	8.56
Office Supplies	8.56	4.54	4.02	0.8855	8.56	200.99
Furniture	200.99	168.82	32.17	0.1606	180.495	34.54
Office Supplies	34.54	17.96	16.58	0.9232	27.612	34.54
Office Supplies	34.54	17.96	16.58	0.9232	27.612	120.99
Furniture	120.99	108.88	12.11	0.1112	102.8415	120.99
Office Supplies	120.99	108.88	12.11	0.1112	102.8415	6.48
Office Supplies	6.48	3.37	3.11	0.9228	4.48	58.14
Furniture	58.14	44.19	13.95	0.3157	29.07	9.49
Office Supplies	9.49	5.03	4.46	0.8867	7.592	

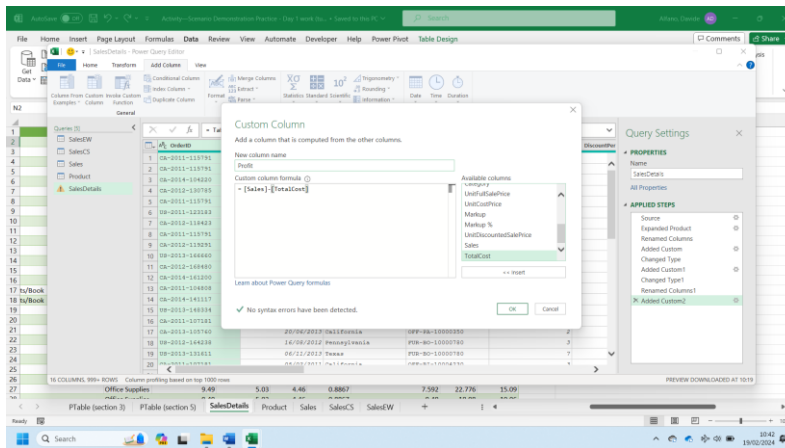
- Create a new column, [Sales], calculated as [UnitDiscountedSalePrice] × [Quantity].



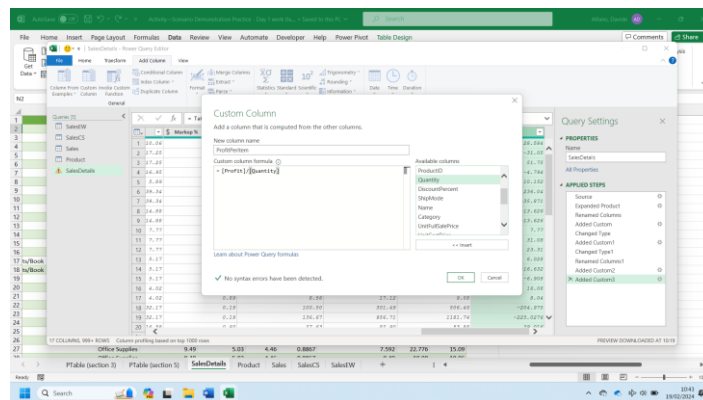
- Rename the [CostPrice] column 'UnitCostPrice'. Create a new column, [TotalCost], calculated as [UnitCostPrice] × [Quantity]. Round and format this column as you deem appropriate.



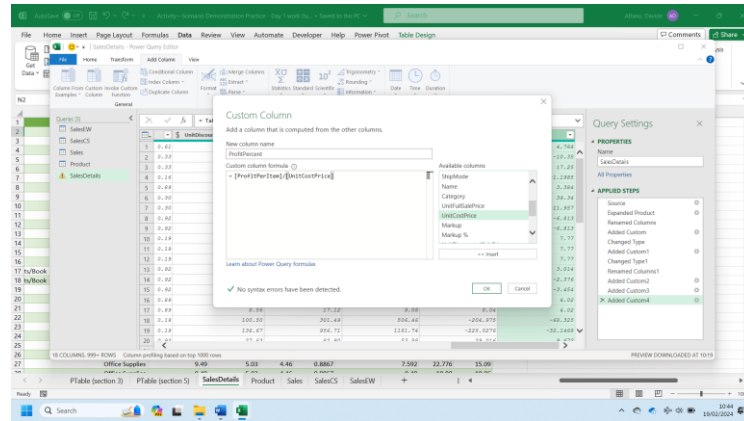
- Create a new column, [Profit], calculated as [Sales] – [TotalCost].



- Create a new column, [ProfitPerItem], and populate it appropriately. Round this column as you deem appropriate.



- Create a new column, [ProfitPercent], calculated as [ProfitPerItem] / [UnitCostPrice].



Part 7 – Analysis

- Calculate the sum of [TotalCost], the sum of [Sales], and the sum of [Profit].

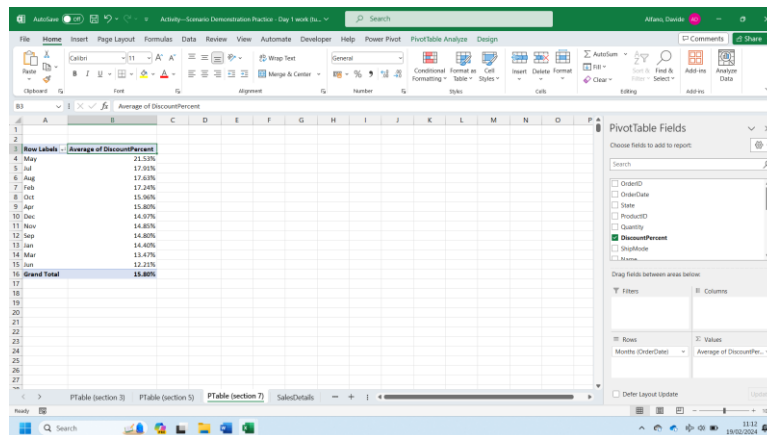
A possible way to determine the same is using the function Autosum() in Excel under the tab Formulas.

	Category	UnitFullSalePrice	UnitCostPrice	Markup	Markup %	UnitDiscountedSalePrice	Sales	TotalCost	Profit	ProfitPerItem	ProfitPercent
1879	Office Supplies	7.78	4.38	3.5	0.8178	7.78	38.9	21.4	17.5	3.5	0.8178
1880	Office Supplies	20.48	14.95	5.53	0.3699	20.48	53.44	44.85	16.59	5.53	0.3699
1881	Office Supplies	6.88	3.72	3.16	0.8495	6.88	27.52	14.88	12.64	3.16	0.8495
1882	Furniture	254.99	204.63	50.36	0.099	254.99	639.96	428.52	41.44	30.36	0.099
1883	Office Supplies	30.53	16.49	14.04	0.8514	30.53	61.06	32.08	28.98	14.04	0.8514
1884	Technology	39.37	48.68	10.69	0.2196	39.37	178.11	146.04	32.07	10.69	0.2196
1885	Office Supplies	303.25	201.54	101.71	0.3489	303.25	1009.75	784.62	305.13	101.71	0.3489
1886	Office Supplies	6.94	4.44	2.5	0.5631	6.94	34.7	22.2	12.5	2.5	0.5631
1887	Office Supplies	8.43	4.55	3.88	0.8527	8.43	33.72	18.2	15.52	3.88	0.8527
1888	Office Supplies	4.99	2.64	2.35	0.8902	4.99	14.97	7.92	7.05	2.35	0.8902
1889	Technology	17.48	10.49	6.99	0.6663	17.48	104.88	62.94	41.94	6.99	0.6663
1890	Furniture	7.38	5.24	2.14	0.4084	7.38	24.76	12.48	4.28	2.14	0.4084
1891	Furniture	348.21	268.12	80.09	0.2987	348.21	974.988	1072.48	-67.492	-24.373	-0.0909
1892	Office Supplies	181.46	128.84	52.62	0.4084	181.46	544.38	386.52	157.86	52.62	0.4084
1893	Technology	9.78	6.16	3.62	0.5677	9.78	29.34	18.48	10.86	3.62	0.5677
1894	Furniture	19.99	15.39	4.6	0.2989	19.99	99.95	76.95	23	4.6	0.2989
1895	Furniture	7.96	4.46	3.5	0.7948	7.96	15.92	8.92	7	3.5	0.7948
1896	Furniture	303.76	255.16	48.6	0.1905	303.76	607.52	530.32	77.2	48.6	0.1905
1897	Office Supplies	165.2	122.25	42.95	0.3513	165.2	264.32	244.5	19.82	9.91	0.0811
1898	Technology	120	63.6	56.4	0.8868	120	127.2	7.2	116	56.4	0.5566
1899	Furniture	8.75	5.08	3.67	0.7224	8.75	21	15.24	5.76	3.67	0.7224
1900							488993	430139	58859		

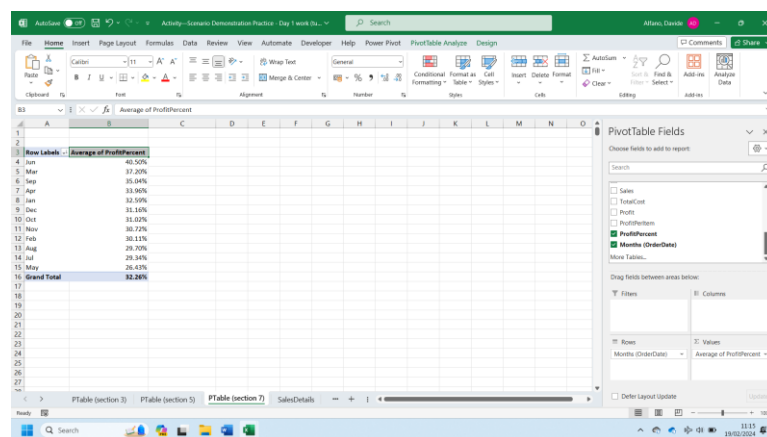
- Verify that sum of [Profit] is equal to sum of [Sales]- sum of [TotalCost] as estimated before through the Autosum function.

	Category	UnitFullSalePrice	UnitCostPrice	Markup	Markup %	UnitDiscountedSalePrice	Sales	TotalCost	Profit	ProfitPerItem	ProfitPercent
1879	Office Supplies	7.78	4.38	3.5	0.8178	7.78	38.9	21.4	17.5	3.5	0.8178
1880	Office Supplies	20.48	14.95	5.53	0.3699	20.48	53.44	44.85	16.59	5.53	0.3699
1881	Office Supplies	6.88	3.72	3.16	0.8495	6.88	27.52	14.88	12.64	3.16	0.8495
1882	Furniture	254.99	204.63	50.36	0.099	254.99	639.96	428.52	41.44	30.36	0.099
1883	Office Supplies	30.53	16.49	14.04	0.8514	30.53	61.06	32.08	28.98	14.04	0.8514
1884	Technology	39.37	48.68	10.69	0.2196	39.37	178.11	146.04	32.07	10.69	0.2196
1885	Office Supplies	303.25	201.54	101.71	0.3489	303.25	1009.75	784.62	305.13	101.71	0.3489
1886	Office Supplies	6.94	4.44	2.5	0.5631	6.94	34.7	22.2	12.5	2.5	0.5631
1887	Office Supplies	8.43	4.55	3.88	0.8527	8.43	33.72	18.2	15.52	3.88	0.8527
1888	Office Supplies	4.99	2.64	2.35	0.8902	4.99	14.97	7.92	7.05	2.35	0.8902
1889	Technology	17.48	10.49	6.99	0.6663	17.48	104.88	62.94	41.94	6.99	0.6663
1890	Furniture	7.38	5.24	2.14	0.4084	7.38	24.76	12.48	4.28	2.14	0.4084
1891	Furniture	348.21	268.12	80.09	0.2987	348.21	974.988	1072.48	-67.492	-24.373	-0.0909
1892	Office Supplies	181.46	128.84	52.62	0.4084	181.46	544.38	386.52	157.86	52.62	0.4084
1893	Technology	9.78	6.16	3.62	0.5677	9.78	29.34	18.48	10.86	3.62	0.5677
1894	Furniture	19.99	15.39	4.6	0.2989	19.99	99.95	76.95	23	4.6	0.2989
1895	Furniture	7.96	4.46	3.5	0.7948	7.96	15.92	8.92	7	3.5	0.7948
1896	Furniture	303.76	255.16	48.6	0.1905	303.76	607.52	530.32	77.2	48.6	0.1905
1897	Office Supplies	165.2	122.25	42.95	0.3513	165.2	264.32	244.5	19.82	9.91	0.0811
1898	Technology	120	63.6	56.4	0.8868	120	127.2	7.2	116	56.4	0.5566
1899	Furniture	8.75	5.08	3.67	0.7224	8.75	21	15.24	5.76	3.67	0.7224
1900							488993	430139	58859		

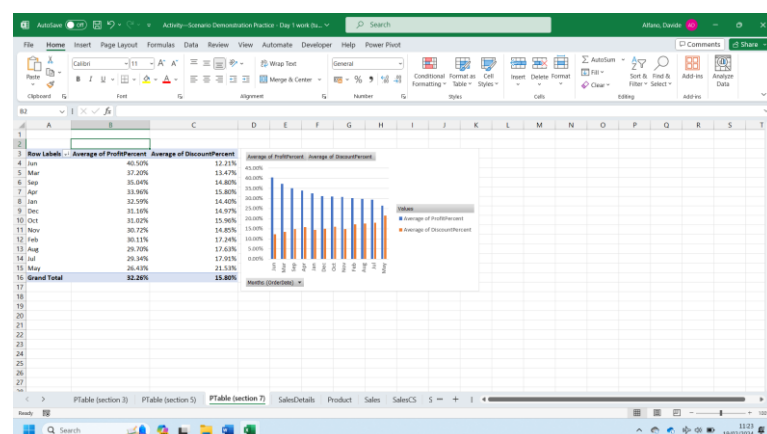
- In which month of the year is the mean [DiscountPercent] highest? You can use a Pivot Table to estimate the required information, as shown below:



- In which month of the year is the mean [ProfitPercent] highest? Again, a Pivot Table as shown below will help you answering the question.

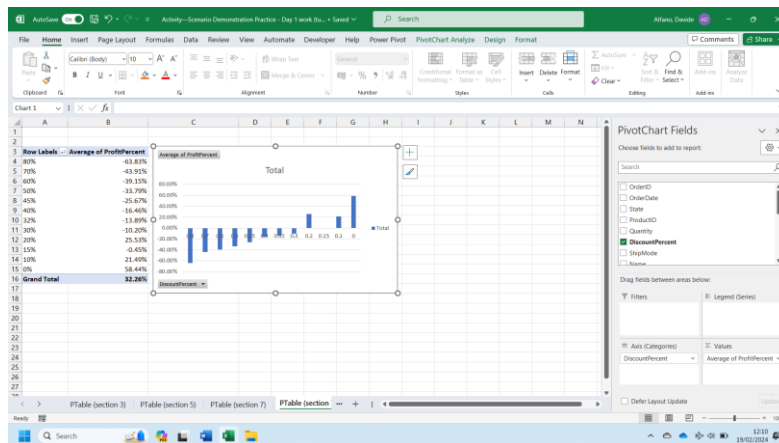


- If you were to plot the mean [ProfitPercent] vs. [DiscountPercent], what do you predict the trend would look like?



Using the average monthly values estimated earlier, the highest the applied discount the lowest will be the profit.

- Calculate the mean [ProfitPercent] for each distinct value of [DiscountPercent]. Plot a suitable chart. Briefly describe the trend.



Again, the evidence is that the highest the applied discount, the lowest will be the average profit obtained.

- For each [State], calculate the sum of [Quantity] and the sum of [Profit].

A Pivot Table as shown below may help in visualizing the required information

State	Sum of Profit	Sum of Quantity
Alabama	413.33	33.00
Arizona	845.70	208.00
Arkansas	1,092.04	22.00
California	13,557.15	1,265.00
Colorado	22.65	149.00
Connecticut	603.09	41.00
Delaware	1,236.80	67.00
Florida	89.86	232.00
Georgia	3,921.35	217.00
Idaho	146.01	19.00
Illinois	889.80	295.00
Indiana	1,515.52	124.00
Iowa	122.22	25.00
Kansas	68.23	5.00
Kentucky	2,984.84	55.00
Louisiana	120.91	35.00
Maryland	152.64	42.00
Massachusetts	1,084.60	65.00
Michigan	10,322.56	230.00
Minnesota	6,148.81	85.00
Mississippi	120.27	23.00
Missouri	546.62	35.00
Montana	1,455.60	13.00
Nebraska	137.65	16.00
Nebraska	137.65	16.00

- How many [State]s have a negative sum of [Profit]?

Appropriately filtering the Pivot Table, as defined below, you will determine the states which have a total profit below zero.

The screenshot shows an Excel window with a PivotTable. The PivotTable has 'State' as the Row Labels and 'Sum of Profit' and 'Sum of Quantity' as the Values. A 'Value Filter (State)' dialog box is open, showing 'Sum of Profit' and a filter criteria of 'is less than 0'. The PivotTable Fields task pane on the right shows 'State' in the Rows area and 'Sum of Profit' and 'Sum of Quantity' in the Values area.

State	Sum of Profit	Sum of Quantity
New York	17,770.84	901.00
California	13,557.15	1,265.00
Michigan	10,322.56	235.00
Virginia	6,405.98	225.00
Minnesota	6,148.81	85.00
Washington	5,791.71	2.00
Georgia	3,912.35	140.00
Kentucky	2,884.84	13.00
New Jersey	2,557.99	13.00
Wisconsin	1,883.07	13.00
Vermont	1,528.65	13.00
Indiana	1,315.52	13.00
Montana	1,455.90	13.00
Delaware	1,236.80	67.00
Oklahoma	1,118.86	59.00
Arkansas	1,092.04	22.00
Massachusetts	1,084.60	65.00
Rhode Island	652.19	48.00
Connecticut	603.09	41.00
Missouri	546.62	35.00
Utah	539.01	37.00
Nevada	471.49	21.00
Alabama	413.33	33.00
Maryland	332.64	42.00
Grand Total	26,170.75	2,892.00

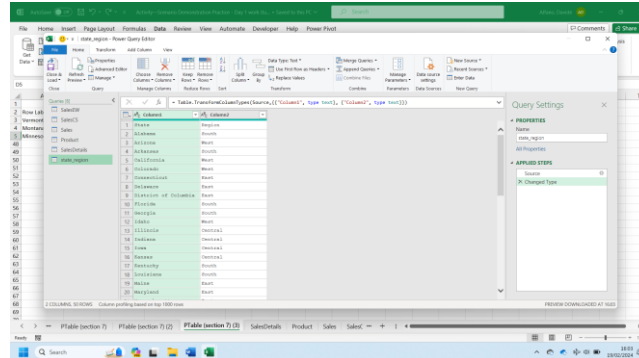
- For each [State], calculate (sum of [Profit]) / (sum of [Quantity]) to obtain the mean profit per item for that [State]. What are the top 3 [State]s by mean profit per item?

The screenshot shows an Excel window with a PivotTable. The PivotTable has 'State' as the Row Labels and 'Sum of Profit' and 'Sum of Quantity' as the Values. A calculated field is added to the PivotTable, showing the formula $\text{Sum of Profit} / \text{Sum of Quantity}$ in the 'Sum of Profit / Sum of Quantity' column. The PivotTable Fields task pane on the right shows 'State' in the Rows area and 'Sum of Profit' and 'Sum of Quantity' in the Values area.

State	Sum of Profit	Sum of Quantity	Sum of Profit / Sum of Quantity
Vermont	1528.65	13	117.59
Montana	1455.9	13	111.97
Minnesota	6148.81	85	72.34

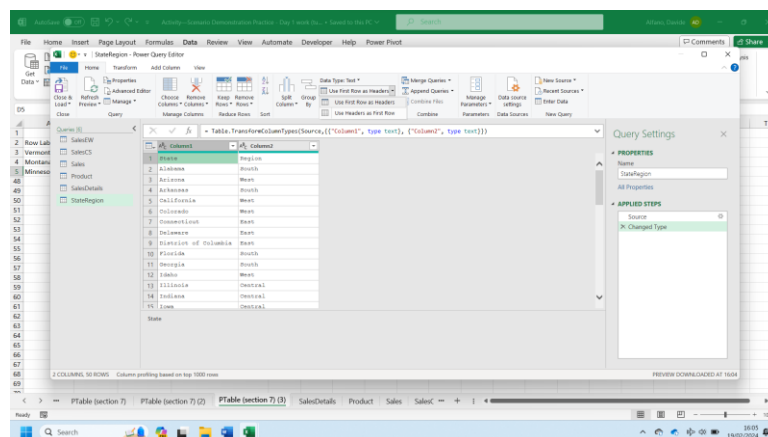
Part 8

- In the 'data' folder, locate the tab-delimited text file 'state_region.txt'. Import this file. Name the table 'StateRegion'.

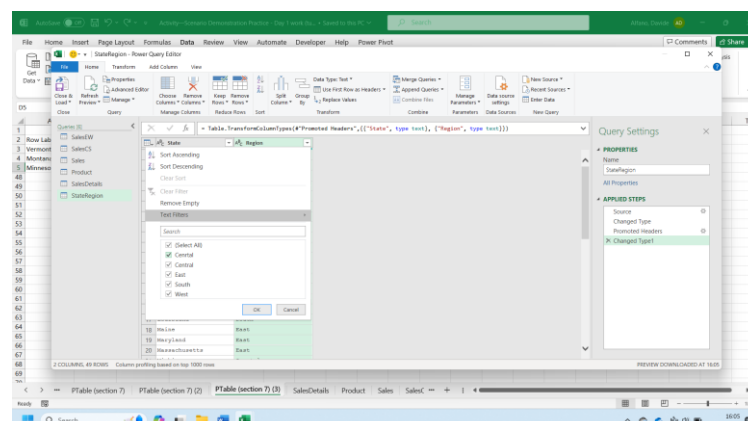


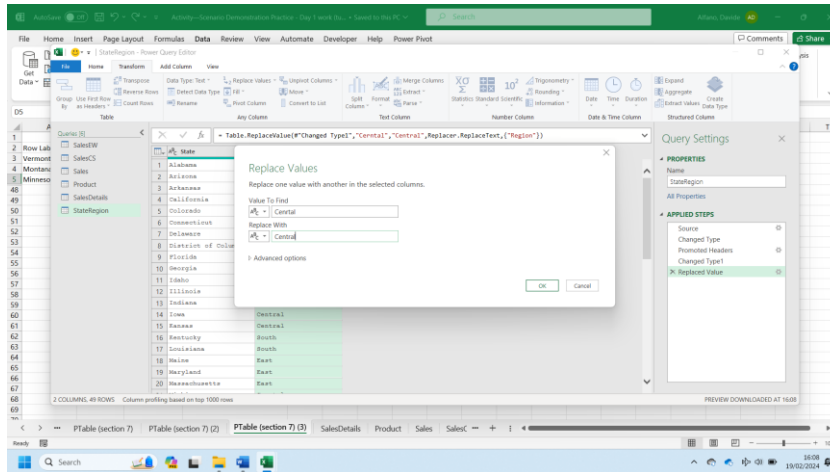
- Find and correct the misspelling in the [Region] column. Verify that the only distinct regions are now 'East', 'West', 'Central' and 'South'.

Firstly, select the first row as header:

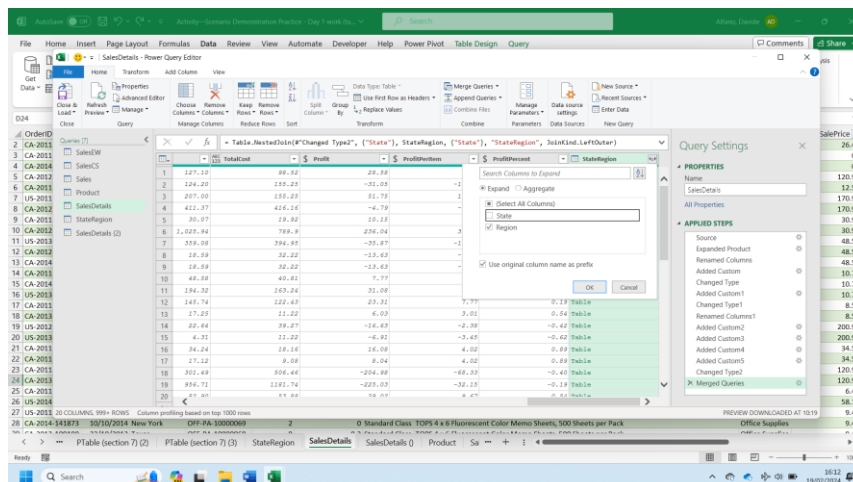
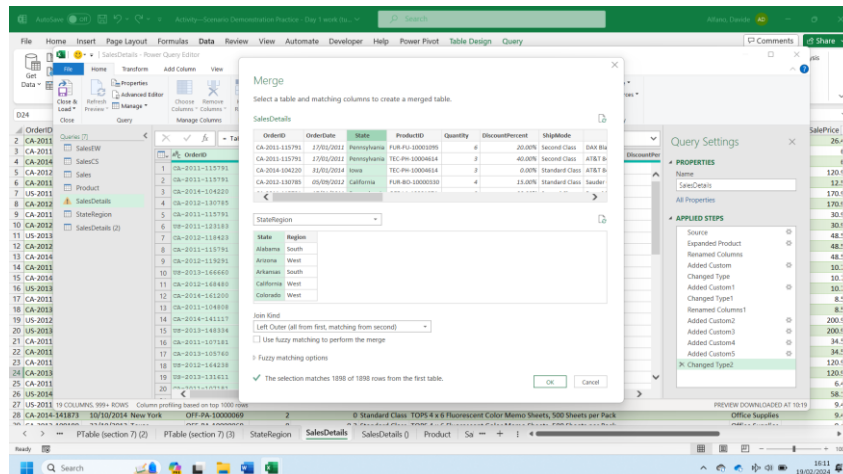


The word central has been misspelt. Use replace values to fix this:





- Modify the SalesDetails table by merging the StateRegion table to it on a suitable join column. Use a left join. Bring in only the [Region] column from the StateRegion table.



Finale, save the SalesDetails table in the following formats : CSV, Text, Excel workbook.

Part 9 - Analysis

- Calculate the total [Profit] by [Region] and [Category]. ,
 - For which combination of {[Region], [Category]} is total [Profit] highest?
 - For which {[Region], [Category]} is total [Profit] negative?

You may refer to the following Pivot Table to identify the answers to both questions:

Region	Office Supplies	Technology	Furniture	Grand Total
Central	1,327	9,426	4,939	15,692
East	1,239	7,470	2,138	10,847
South	2,353	6,333	5,003	13,689
West	1,477	8,267	5,702	15,446
Grand Total	6,406	31,497	17,782	55,685

- West - Technology
 - Central – Furniture
- Calculate the sum of [Quantity] by [Region] and [ShipMode]. Display the sums as a percentage of the total [Quantity] for each [Region].
In which [Region] is the percentage of items delivered First Class higher than the percentage of items delivered Second Class?
Firstly, let's create a Pivot Table with the appropriate fields:

Region	First Class	Second Class	Standard Class	Grand Total
Central	223	79	251	553
East	365	62	439	866
South	109	19	281	409
West	379	128	368	875
Grand Total	1,076	188	1,309	2,573

Then, display the values as percentage of row values:

The screenshot shows an Excel PivotTable with the following data:

Region	First Class	Second Class	Standard
Central	215	78	351
East	361	82	478
South	139	99	203
West	379	110	364
Grand Total	1,094	309	1,396

The context menu for the PivotTable is open, showing the following options:

- Copy
- Format Cells...
- Number Format...
- Refresh
- Sort
- Remove "Sum of Quantity"
- Summarize Values By
- Show Values As
- Show Details
- Value Field Settings...
- PivotTable Options...
- Hide Field List

The 'Show Values As' option is selected, displaying a submenu with the following options:

- By Calculation
- % of Grand Total
- % of Column Total
- % of Row Total
- % of...
- % of Parent Row Total
- % of Parent Column Total
- % of Parent Total...
- Difference From...
- % Difference From...
- Running Total In...
- % Running Total In...
- Rank Smallest to Largest...
- Rank Largest to Smallest...
- Index
- More Options...

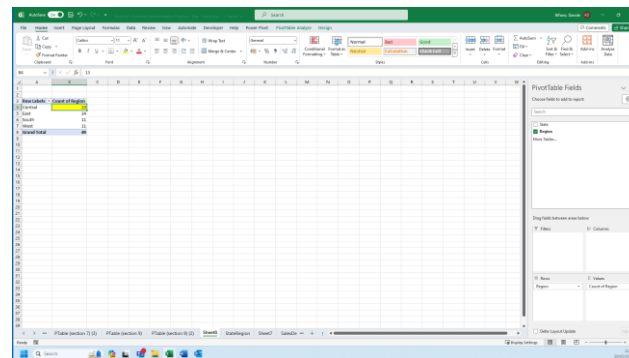
The 'By Calculation' option is selected, displaying a tooltip that reads: "Display all the values in each row as a percentage of the total for the row."

West is the only region where the percentage of First Class Shipping is higher than Second Class Shipping.

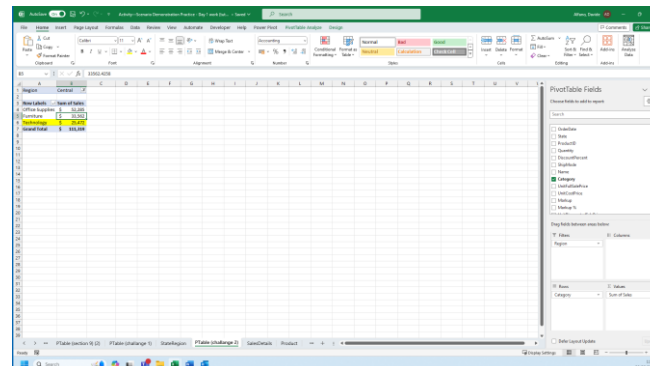
Challenge

- Find the product with the highest [UnitFullSalePrice] within the [Category] with the lowest sum of [Sales] within the [Region] with the second highest number of [State]s.

Firstly define which is the region with the second highest number of states creating a pivot table based on StateRegion:



Secondly, within Central Region, let's define the Category with the lowest sum of Sales using a pivot table based on SalesDetails:



Let's now find the product with the highest UnitFullSalePrice within the Technology category

